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PRODUCT CONFIGURATION ADMINISTRATION GUIDE

ENTERPRISE SOLUTIONS → STORE ADMINISTRATION

DELIVERED TO / ACCEPTED BY



3D GALAXY E-COMMERCE & CUSTOM MANUFACTURING PLATFORM

PRODUCT CONFIGURATION GUIDE

**COMPLETE OPERATIONAL REFERENCE, ARCHITECTURE & FIELD SPECIFICATION
MANUAL**

Professional Client-Facing Manual featuring Clean Native Application Screenshots, Exhaustive Field References, Category Architecture, Cartesian Variant Combination Matrix, Bundle Builder, Dynamic IST Delivery Date Engine, and Pre-Publish Validation Gates.

DELIVERED BY

AJR Digital Hub Team

Enterprise Digital Solutions & Architecture

DELIVERED TO / ACCEPTED BY

3D Galaxy Team

Catalog Operations & Store Administration

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Commercial in Confidence

Clean Application Visuals (Zero Tour Annotations)

Complete Operational Reference, Architecture & Field Specification Manual

DOCUMENT CONTROL & GOVERNANCE

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DOCUMENT REVISION HISTORY

VERSION	RELEASE DATE	AUTHOR / ENTITY	SUMMARY OF OPERATIONAL SCOPE
1.0	Sep 18, 2026	AJR Digital Hub Engineering	Initial extraction of data models, Angular signals, reactive form controls, and validation rules.
1.5	Sep 20, 2026	AJR Digital Hub Solutions	Dropdown documentation, Cartesian matrix permutations, and dynamic delivery date formulas.
2.0	Sep 21, 2026	AJR Digital Hub Delivery Team	Complete Clean Screenshot Edition: Removed all tour callouts, numbered circles, arrows, and overlay boxes. Replaced with pristine, native application screenshots and comprehensive field reference tables, option matrices, and outside operational callout boxes.

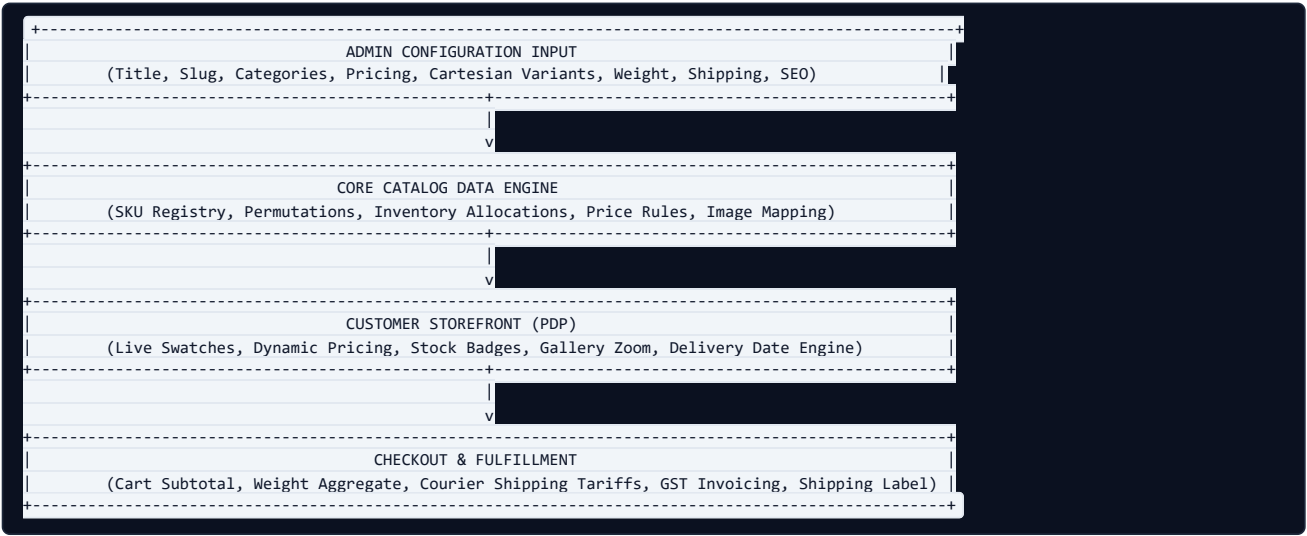
1. DOCUMENT PURPOSE & EXECUTIVE SUMMARY

The **3D Galaxy Product Configuration Administration Guide** is an exhaustive, professional operational manual created specifically for the 3D Galaxy catalog administration and fulfillment team.

Unlike a temporary interactive tour or walkthrough, this document serves as a permanent reference manual. Every visual element inside this guide represents the **authentic, clean application user interface** exactly as experienced inside the 3D Galaxy Admin Portal—with zero overlay graphics, zero tour markers, and zero obscuring callout boxes.

CORE DOCUMENTATION PARADIGM

- 1. **Clean Application Visuals:** Every screenshot provides an unadulterated, high-resolution view of the application UI.
- 2. **Exhaustive Field References:** Every control, input box, toggle, and dropdown is tabulated and explained directly below the corresponding figure.
- 3. **Verified Option Catalogs:** All dropdown menus, display types, and selection modes reflect the exact code specifications within the platform.
- 4. **Storefront Impact Analysis:** Clear explanations of how administrative inputs alter customer product detail pages (PDP), cart subtotals, weight-based freight charges, and invoice generation.
- 5. **Quality Gates & Operational Safeguards:** Actionable validation checklists and callout boxes placed safely outside screenshots to prevent administrative misconfiguration.



2. CATALOG ARCHITECTURE & REGISTRY CONSOLE

The Catalog Registry console is the operational headquarters where all products, SKUs, inventory counts, dealer rates, and catalog lifecycle actions are monitored, filtered, and initiated.

FIGURE 1.1 — CATALOG REGISTRY CONSOLE

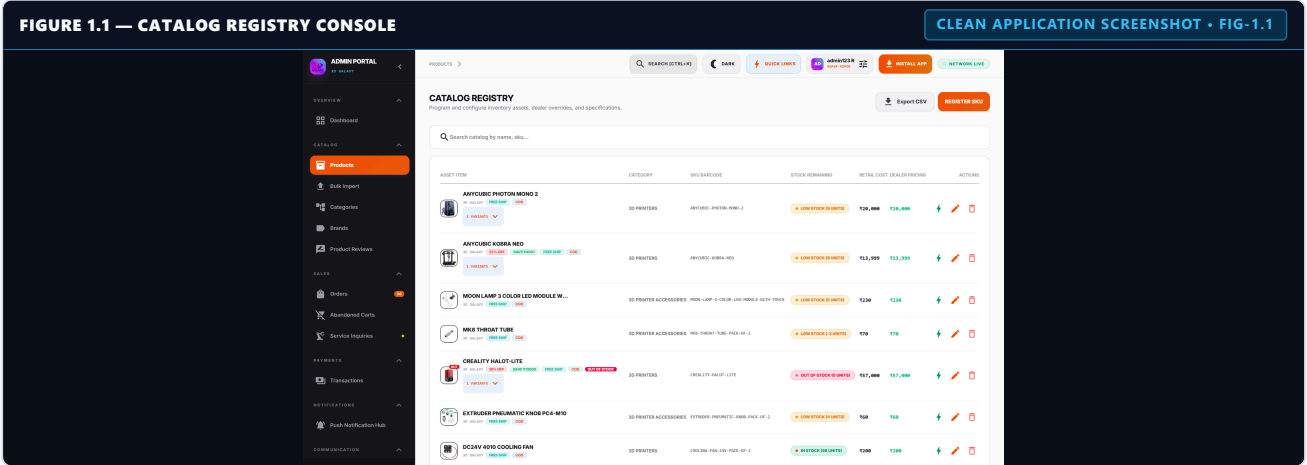


Figure Description: Catalog Registry Console showing active product listings, SKU codes, category assignments, stock status badges, pricing columns, and top-level action controls.

CONTROL REFERENCE

CONTROL / COLUMN	PURPOSE	ADMIN ACTION
+ Add Product	Initiates new product creation workflow	Click to launch blank product configuration editor
Search Input	Real-time filtering across titles and SKUs	Type keywords, part numbers, or unique SKUs to isolate records
Category Filter	Multi-taxonomy drill-down	Filter list by specific product category or department
Stock Status Badge	Real-time stock state indicator	Evaluates physical inventory (In Stock , Low Stock , Out of Stock)
Base Price / MRP	Retail and discounted price display	Quickly audit consumer selling prices across the catalog
Action Menu (...)	Row-level operational commands	Edit product, view live preview, duplicate SKU, or deactivate item

ADMINISTRATOR TIP

ADMIN TIP

Use the Catalog Search filter to audit existing SKUs before creating new products. Maintaining a strict, standardized SKU syntax (e.g., `FIL-PLA-BLK-1KG`) ensures clean inventory tracking and prevents accidental duplicates across the catalog.

3. GENERAL PRODUCT IDENTITY & CLASSIFICATION

The General tab establishes the core data attributes, search identifiers, and public marketing copy for the product.

FIGURE 2.1 — GENERAL PRODUCT IDENTITY

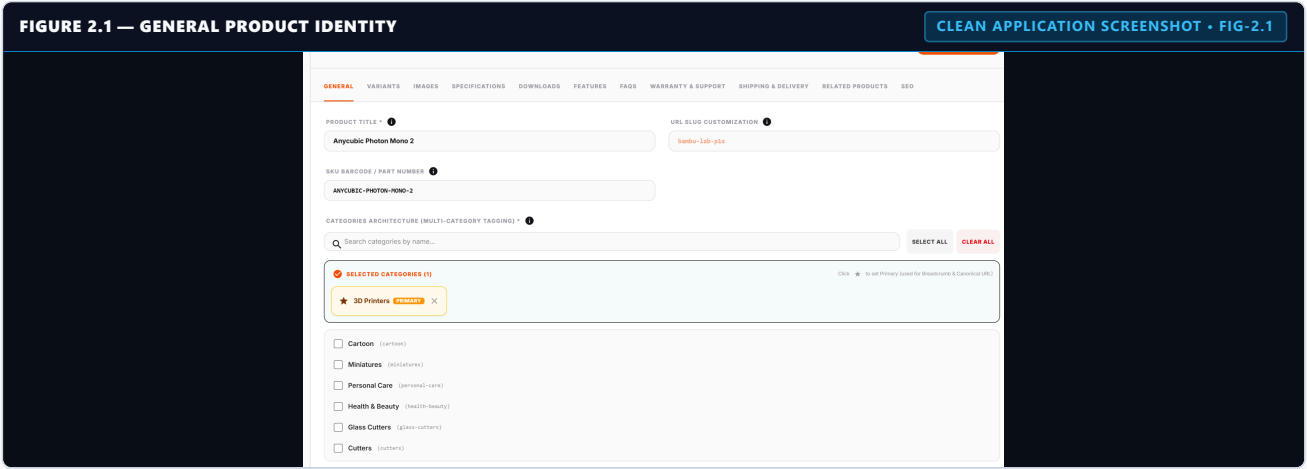


Figure Description: General product configuration screen used to configure the core product title, URL slug, SKU, category associations, and descriptive marketing copy.

FIELD REFERENCE

FIELD	DESCRIPTION	CONFIGURATION
Product Title	Primary customer-facing product name	Enter clear, keyword-rich title (max 255 chars)
URL Slug	Clean, SEO-friendly URI path identifier	Auto-generated from title or custom kebab-case slug
SKU / Barcode	Unique master product stock keeping unit & EAN	Enter alphanumeric internal SKU and scannable barcode
Categories	Hierarchical taxonomy classification	Select one or more relevant category nodes
Short Description	Concise summary displayed on product cards & top PDP	Enter 1–2 high-impact sentences summarizing key benefits
Full Description	Comprehensive technical and marketing overview	Rich text formatting with specifications, benefits, and usage

DETAILED FIELD EXPLANATIONS

- Product Title:** The customer-facing identity of the item. It is automatically propagated to the HTML `<title>` tag, OpenGraph social cards, breadcrumbs, search index tokens, and order line items.
- URL Slug:** The clean URI path appended to `https://3dgalaxy.in/product/`. Must contain only lowercase alphanumeric characters and hyphens (e.g., `creality-ender-3-v3-ke`). Changing the slug on published products breaks legacy search links unless a 301 redirect is provisioned.
- SKU / Barcode:** Internal warehouse inventory identifier and scannable GS1/EAN barcode. If the product has Cartesian variants, this serves as the parent base SKU.
- Short Description:** Rendered immediately below the product title and star rating on the storefront PDP. It also forms the default fallback for search engine meta descriptions.
- Full Description:** Fully featured rich text HTML editor supporting headings, bullet points, technical comparison tables, and embedded diagrams.

CRITICAL SPECIFICATION

IMPORTANT

The URL slug must be strictly unique across the entire database. If a duplicate slug is submitted, the system will reject the update to prevent URL collision and 404 routing conflicts.

4. CATEGORY CONFIGURATION & ARCHITECTURE

Categorization governs how products are discovered through main navigation menus, category pages, breadcrumbs, and multi-faceted search filters.

FIGURE 3.1 — CATEGORY SELECTOR & ARCHITECTURE

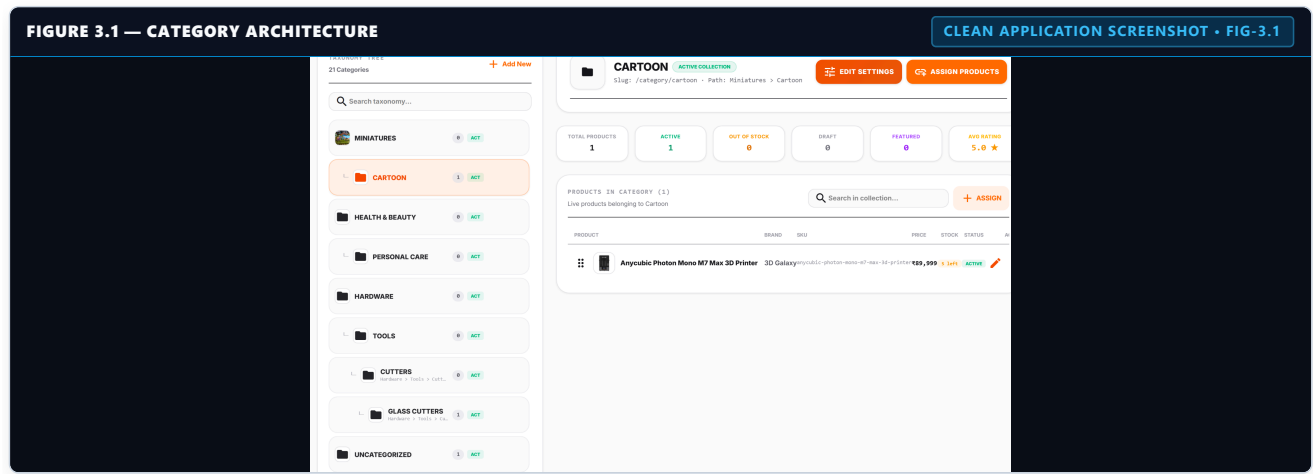


Figure Description: Category selector showing category search bar, category tree checkboxes, bulk selection controls, and selected category tag chips.

CATEGORY ARCHITECTURE

The 3D Galaxy taxonomy engine utilizes a multi-category hierarchical structure allowing a product to belong to multiple catalog categories simultaneously while maintaining a single **Primary Category** for canonical URL routing and breadcrumb navigation.

- **Category Search:** Instantly narrows large category trees by filtering matching parent and child nodes in real time.
- **Category Selection:** Administrators click individual checkboxes to assign products to specific catalog nodes.
- **Multi-Category Selection:** Allows cross-listing items across diverse catalog branches (e.g., listing a 3D printing filament under both *Filaments > PLA* and *Materials > Engineering Consumables*).
- **Select All:** Bulk selects all categories currently visible within the filtered search view.
- **Clear All:** Instantly deselects all active categories, returning the selector to a clean slate.
- **Primary Category:** The definitive parent category used to construct canonical breadcrumb trails (Home > 3D Printers > FDM Printers > [Product]) and Google Structured Data.
- **Selected Category Tags:** Visual chip badges rendering selected categories with instant removal (x) capability.
- **Category Synchronization:** Real-time synchronization ensuring products instantly appear or disappear from storefront category listings upon saving.
- **Category Hierarchy:** Parent-child relationships ensuring that filtering by a parent node automatically surfaces items assigned to its sub-categories.

CONTROL REFERENCE

CONTROL	PURPOSE	ADMIN ACTION
Search	Find categories quickly in dense taxonomy trees	Search by category name or keyword
Checkbox	Select or deselect individual category node	Enable checkbox to associate product
Select All	Select all currently filtered categories in bulk	Click to bulk assign visible categories
Clear All	Remove all selected category assignments	Click to wipe category associations
Primary	Define primary category for canonical routing	Click star or radio icon to designate primary node
Selected Tags	Review and manage active selections	Click 'x' on any chip to remove that category

OPERATIONAL SAFEGUARD

CUSTOMER IMPACT

Failing to assign at least one category will cause the product to become an "orphan item." It will be completely invisible on storefront category browsing pages, mega-menus, and faceted navigation filters, discoverable only by direct URL.

5. BRAND ALLIANCE & TAXONOMY TAGS

FIGURE 4.1 — BRAND ALLIANCE & TAXONOMY

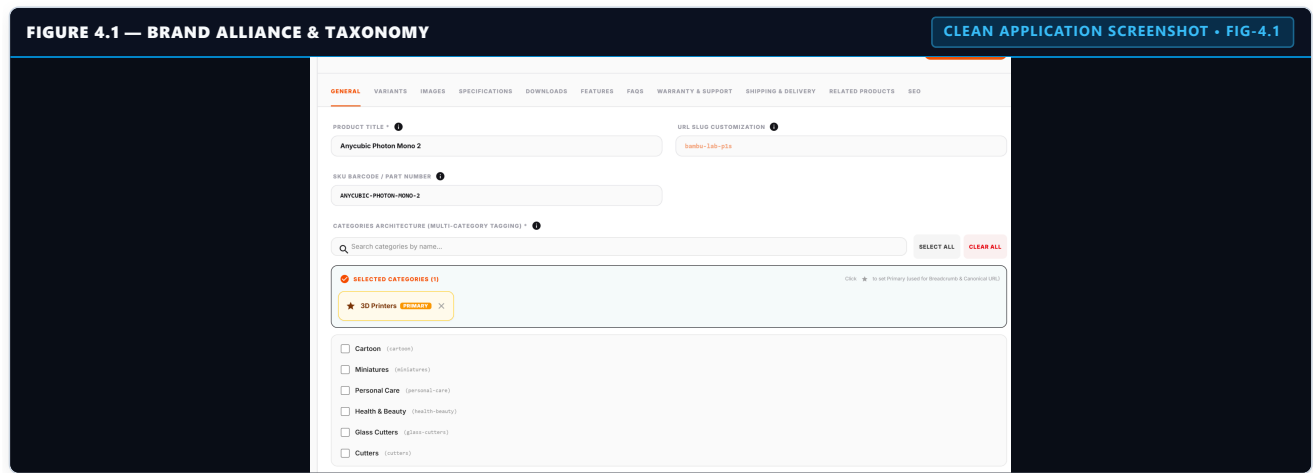


Figure Description: Brand alliance dropdown and searchable taxonomy tag interface for manufacturer association and storefront filtering.

FIELD REFERENCE

FIELD	DESCRIPTION	CONFIGURATION
Brand Alliance	Manufacturer or partner brand association	Select registered brand from dropdown
Taxonomy Tags	Free-form semantic search keywords	Add comma-separated search terms and tags
Model / Series	Hardware revision or filament series	Enter specific manufacturer hardware generation
Compatibility Flags	Associated printer hardware ecosystems	Select supported 3D printer platforms

6. CORE PRICING ARCHITECTURE

Pricing directly controls storefront retail presentation, strike-through promotional displays, dealer rate cards, and cart tax computations.

FIGURE 5.1 — CORE PRODUCT PRICING

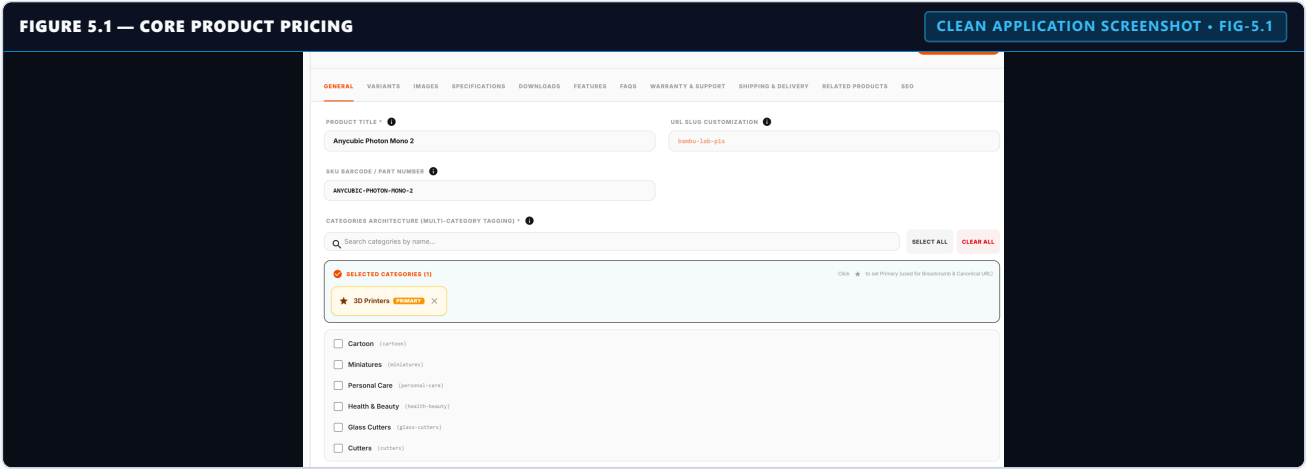


Figure Description: Core product pricing configuration interface displaying MRP, Retail Sale Price, Authorized Dealer Price, and GST tax settings.

FIELD REFERENCE

FIELD	DESCRIPTION	CONFIGURATION
MRP (Maximum Retail Price)	Manufacturer reference price (strike-through)	Enter statutory maximum retail price
Retail Sale Price	Actual selling price charged to standard B2C customers	Enter active selling price (must be ≤ MRP)
Dealer Price (B2B)	Discounted rate for verified B2B/wholesale accounts	Enter wholesale rate unlocked upon B2B login
Tax Rate (GST)	Applicable Goods and Services Tax percentage	Select statutory GST slab (e.g., 18%)
Tax Inclusive Toggle	Determines if entered prices include GST	Toggle ON if price includes tax; OFF for net + tax

CRITICAL SPECIFICATION

VALIDATION RULE

The **Retail Sale Price** must always be less than or equal to the **MRP**. If the Sale Price exceeds MRP, the platform blocks saving with a validation error to comply with consumer protection regulations.

7. BASE INVENTORY MANAGEMENT & STOCK

FIGURE 6.1 — BASE INVENTORY & STOCK

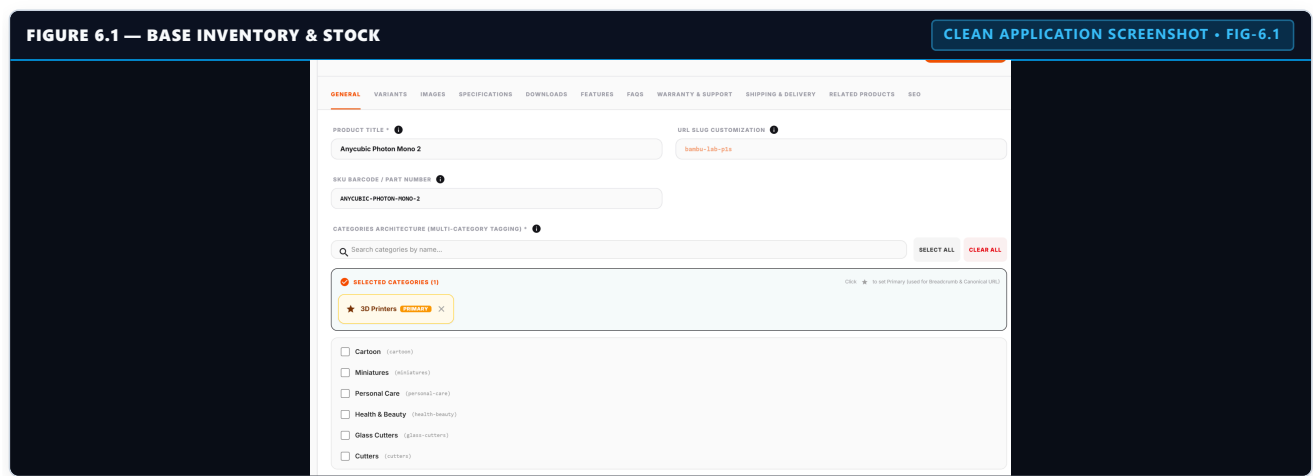


Figure Description: Base inventory management showing master stock counters, low stock warning thresholds, backorder toggles, and inventory tracking flags.

FIELD REFERENCE

FIELD	DESCRIPTION	CONFIGURATION
Track Inventory	Enables automated real-time stock deduction	Toggle ON to track quantities against sales
Current Stock Quantity	Physical units available in fulfillment warehouse	Enter verified on-hand inventory count
Low Stock Threshold	Threshold triggering automated restock alerts	Enter warning number (e.g., 5 units)
Allow Backorders	Enables customer purchases when stock reaches zero	Select Do not allow , Allow , or Notify customer
Stock Status Override	Manual state override for catalog display	Select In Stock , Out of Stock , or On Backorder

8. VARIANT CONFIGURATION ENGINE

The 3D Galaxy Variant Engine transforms complex, multi-option items (such as filaments with multiple colors, spool weights, and bundle packs) into streamlined customer selection interfaces.

FIGURE 7.1 — VARIANT GROUP CONFIGURATION

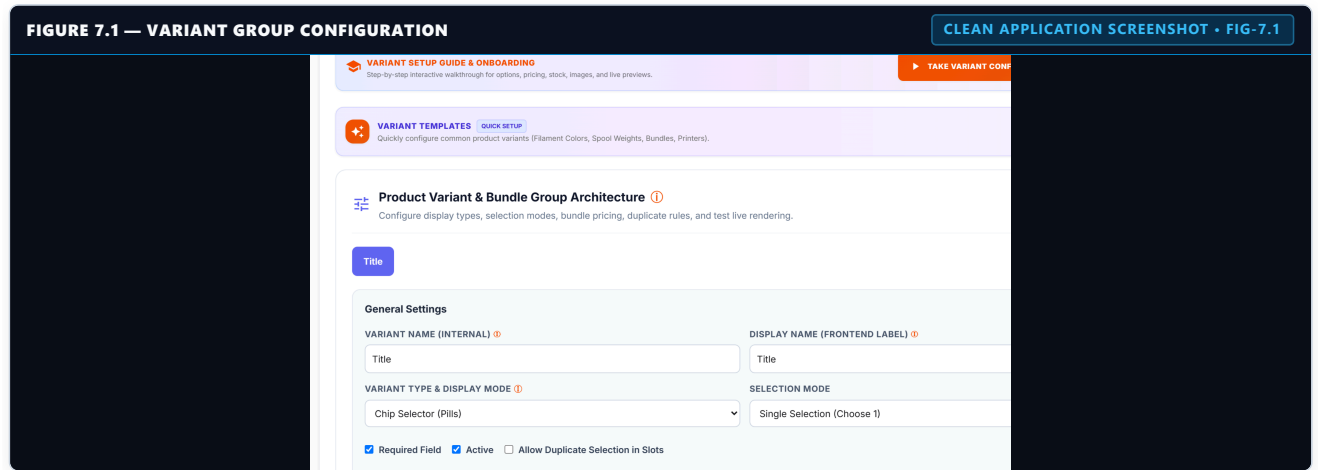


Figure Description: Variant group header configuration showing group naming, display name, display type selector, selection mode, and operational toggles.

VARIANT GROUP

Every variant dimension (e.g., "Color", "Weight", "Nozzle Size") is governed by a dedicated **Variant Group**:

- **Variant Name:** Internal technical identifier used in database keys and Cartesian matrix generation (e.g., `filament_color`).
- **Display Name:** Customer-facing label rendered on the storefront PDP (e.g., "Select Filament Color").
- **Variant Type:** Classification of variant attributes (Attribute, Material, Dimension, Pack Size).
- **Display Mode:** Visual styling employed to render option selectors on the storefront (Chips, Dropdown, Swatches).
- **Selection Mode:** Logical rules governing how customers make selections (Single, Bundle, Weight-based, Multiple).
- **Required Field:** When enabled, customers must select an option from this group before clicking "Add to Cart".
- **Active:** Master toggle enabling or disabling this variant group without deleting configured data.
- **Duplicate Selection:** In multi-selection modes, governs whether a customer can select the same variant item multiple times.

FIELD REFERENCE

FIELD	DESCRIPTION	CONFIGURATION
Variant Group Name	Internal technical reference identifier	Enter concise identifier (e.g., <code>spool_weight</code>)
Display Name	Storefront customer-facing label	Enter clear user prompt (e.g., <code>Choose Spool Weight</code>)
Display Type	Visual presentation UI component	Select one of 11 verified display components
Selection Mode	Cart selection and pricing logic	Select one of 6 verified operational modes
Required	Mandates selection before cart addition	Toggle ON to enforce required choice
Active	Enables or suspends group in live storefront	Toggle ON to publish; OFF to draft/hide

9. VARIANT DISPLAY TYPES

The 3D Galaxy platform supports **11 verified display types** implemented directly within the storefront presentation engine.

FIGURE 8.1 — VARIANT DISPLAY TYPES

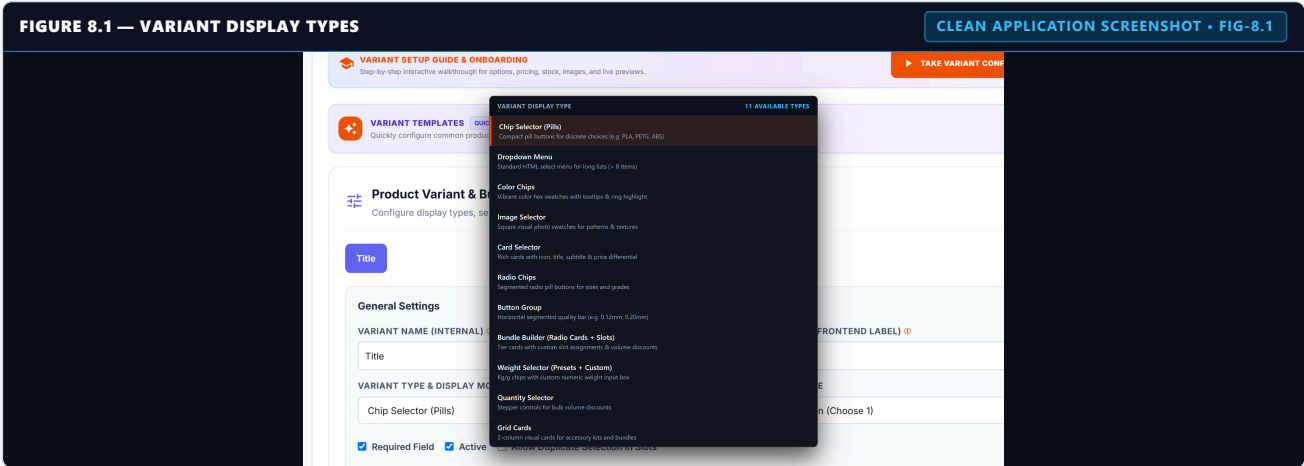


Figure Description: Clean application screenshot showing the Variant Display Types dropdown menu displaying all 11 supported storefront presentation modes.

AVAILABLE OPTIONS REFERENCE TABLE

DISPLAY TYPE	DESCRIPTION	STOREFRONT UI BEHAVIOR	RECOMMENDED USE CASE
Bundle Builder	Interactive tier cards with customizable variant slots	Renders multi-item slot selector with bundle discounts	"Buy 2 Save 10%", "Build 4-Spool Pack"
Weight Selector	Presets and custom weight inputs	Displays weight chips with numeric kg/g custom entry	Bulk materials, resin bottles, filament spools
Chip Selector	Clean horizontal pill buttons	Compact clickable pills highlighting selected state	Clothing sizes, technical specs, nozzle diameters
Dropdown Menu	Standard HTML select box	Compact menu expanding on click	Dense option catalogs (50+ printer models)
Image Selector	Visual photo thumbnail chips	Renders mini preview images with active borders	Filament spool swatches, patterned PEI sheets
Card Selector	Rich detailed cards with descriptions	Cards displaying badges, pricing, and specs	Extended warranty tiers, complete printer bundles
Radio Chips	Circular radio buttons with text	Classic radio selector with visible circular target	Exclusive single-choice binary configurations
Color Chips	Circular/square color hex swatches	Vibrant color discs with tooltips and checkmarks	Filament colors, resin dyes, hardware finishes
Button Group	Segmented inline toggle buttons	Connected button bar with instant tab switching	Fast switching between 2–4 primary modes
Quantity Selector	Stepper counter with tiered bulk discounts	[-] [Qty] [+] controls with volume discount alerts	Consumable spare parts, nozzles, screws
Grid Cards	Multi-column visual grid cards	2-column or 3-column photo cards with subtext	Starter kit builders, modular accessory upgrades

10. VARIANT SELECTION MODES

The selection mode defines the mathematical and cart rules used to evaluate customer selections.

FIGURE 9.1 — SELECTION MODE CONFIGURATION

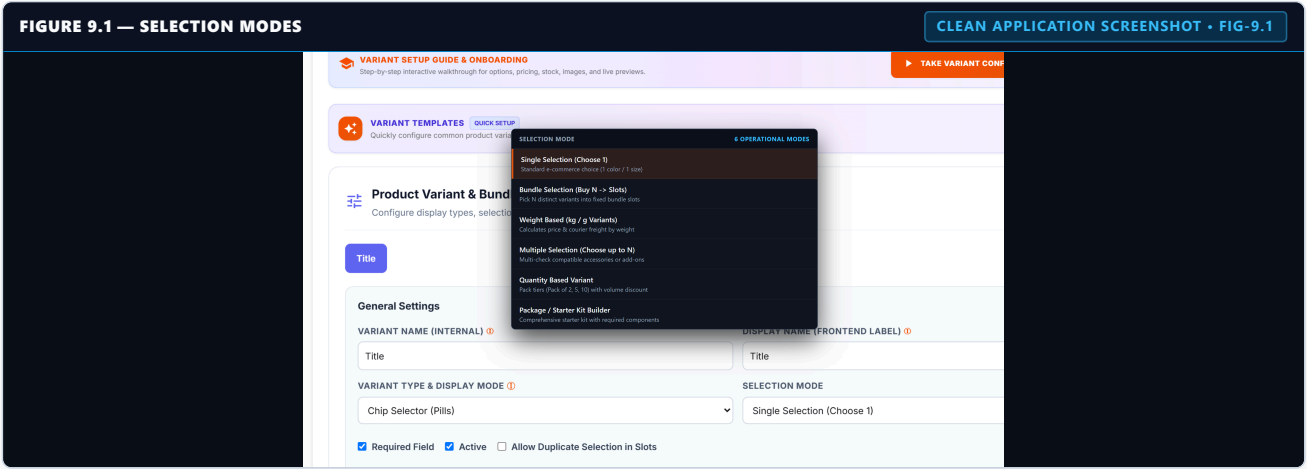


Figure Description: Clean application screenshot showing the Selection Mode dropdown menu displaying all 6 operational selection modes.

AVAILABLE SELECTION MODES REFERENCE TABLE

SELECTION MODE	PURPOSE	CUSTOMER BEHAVIOR	RECOMMENDED USE
Single Selection	Standard e-commerce choice	Customer selects exactly 1 variant option before cart addition	Single filament color, specific machine model
Bundle Selection	Multi-item tiered pack builder	Customer fills N predetermined slots with individual variant choices	4-spool filament variety bundles
Weight Based	Continuous or stepped mass pricing	Customer selects weight preset or types custom weight in kg/g	Industrial resin, bulk plastic pellets
Multiple Selection	Optional add-on accessory check	Customer checks 0, 1, or multiple optional items	Adding extra nozzles, build plates, or tools
Quantity Based	Tiered volume pack discounts	Selecting a pack size automatically updates pricing rules	"Pack of 5", "Pack of 10", "Box of 50"
Package / Starter Kit	Fixed-slot multi-category kit	Customer configures required components across distinct groups	"Complete 3D Printing Starter Kit"

11. VARIANT VALUE REGISTRY & OPTION MODIFIERS

FIGURE 10.1 — VARIANT VALUES REGISTRY

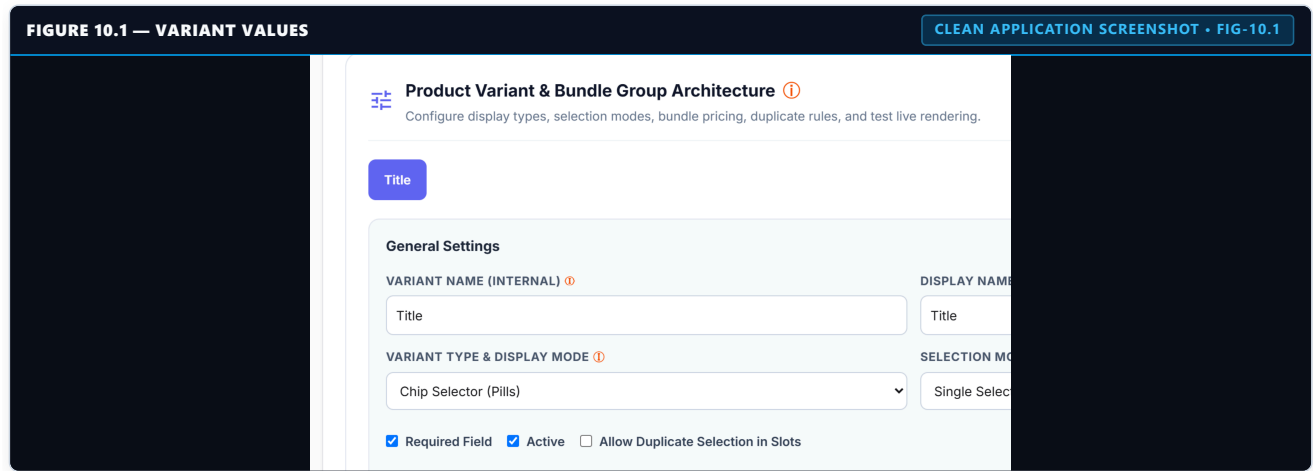


Figure Description: Variant value registry showing configured options, color hex swatches, price adjustments, and weight overrides.

FIELD REFERENCE

FIELD	DESCRIPTION	CONFIGURATION
Option Value Name	Public name of variant option	Enter descriptive name (e.g., Matte Black , 1.75mm)
Hex Color Code	Visual swatch color (for color chips)	Enter 6-digit hex code (e.g., #1A1A1A) or use color picker
Price Adjustment	Differential price added to base price	Enter positive or negative delta (e.g., +250.00)
Weight Override	Specific physical mass for this option	Enter tare/gross weight in grams (e.g., 1000)
Display Order	Sequence of option display on PDP	Enter sort index (1, 2, 3...)

12. VARIANT-SPECIFIC PRICING ARCHITECTURE

When variants have distinct manufacturing costs, the administrator overrides baseline prices directly at the permutation level.

FIGURE 11.1 — VARIANT SPECIFIC PRICING

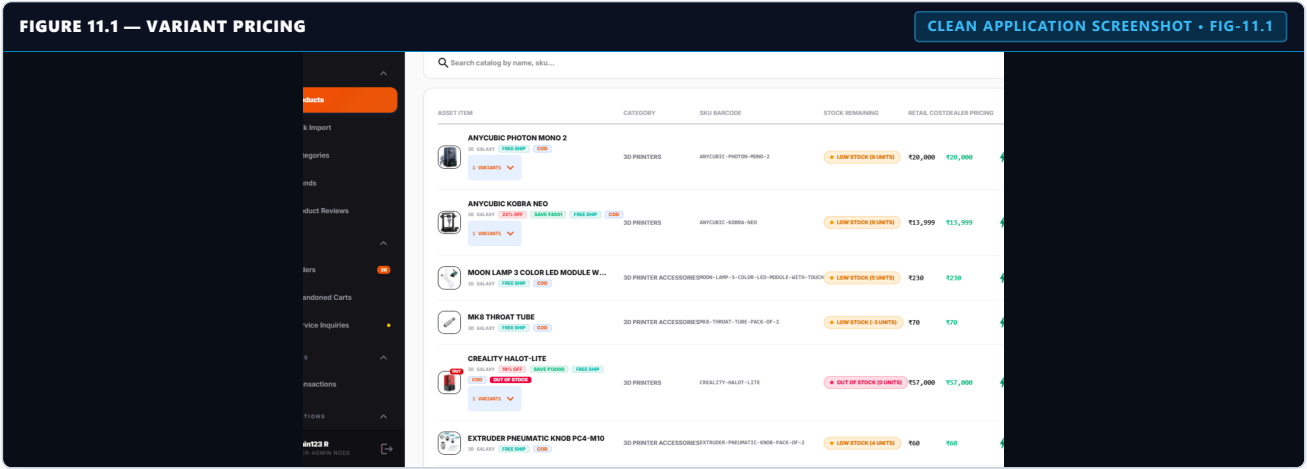


Figure Description: Combination matrix pricing columns displaying MRP, Sale Price, and Dealer Price configurations for each distinct variant permutation.

FIELD REFERENCE

FIELD	DESCRIPTION	CONFIGURATION
Permutation MRP	Maximum Retail Price for this specific variant	Enter statutory MRP for variant (e.g., ₹1,499.00)
Permutation Sale Price	Actual retail price charged for this variant	Enter active selling price (e.g., ₹999.00)
Permutation Dealer Price	Wholesale rate for verified B2B accounts	Enter special wholesale rate (e.g., ₹799.00)
Margin / Discount Badge	Auto-calculated percentage discount	System computes: ((MRP - Sale) / MRP) * 100

13. INVENTORY & STOCK GOVERNANCE

Stock can be managed either globally at the parent product level or granularly at the variant permutation level.

FIGURE 12.1 — VARIANT-LEVEL STOCK INVENTORY

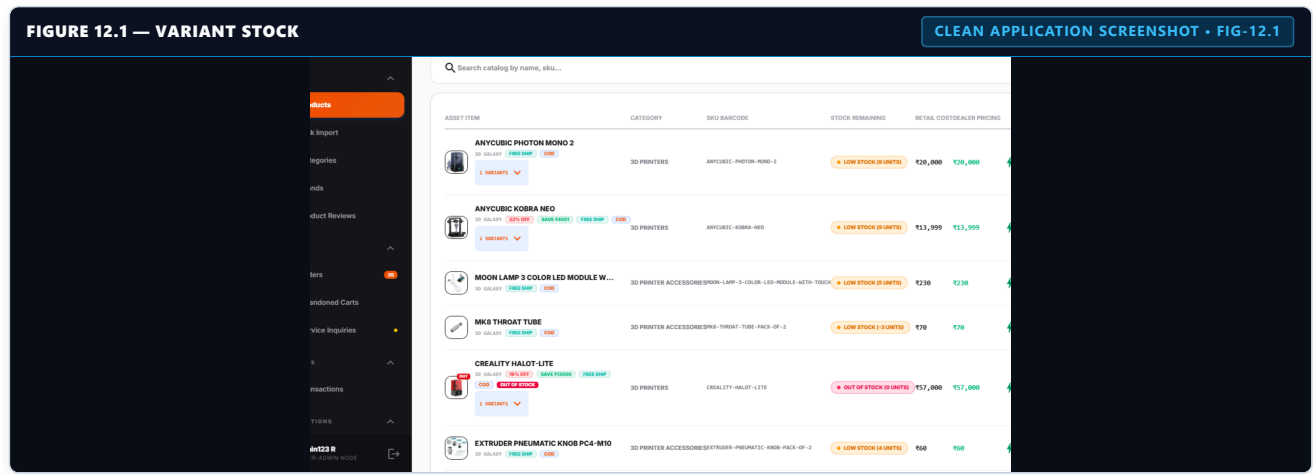


Figure Description: Combination matrix stock inventory interface showing individual warehouse SKU counts, status indicators, and stock management controls.

OPERATIONAL STOCK GOVERNANCE

- Product-Level Stock:** Applied when a product has no variants. A single inventory counter governs the entire product availability.
- Variant-Level Stock:** In products with variant groups, each permutation possesses its own isolated physical stock counter. Selling a "Matte Black / 1kg" spool deducts stock only from that specific variant, leaving "Silk White / 1kg" unaffected.
- Stock Availability Statuses:**
 - In Stock**: Available quantity is greater than zero. Customer can add to cart immediately.
 - Low Stock**: Available quantity is at or below the low stock threshold. Storefront displays an urgent "Only X left!" badge.
 - Out of Stock**: Quantity is zero. Storefront disables the Add to Cart button and presents an out-of-stock badge.
- Out-of-Stock Behavior:** When a customer selects an out-of-stock swatch, the Add to Cart button transitions to "Out of Stock", blocking purchase while keeping other available variants selectable.
- Cart Validation:** The shopping cart continuously polls the real-time stock API. If a customer attempts to increase cart quantity beyond available stock, the system caps the increment and alerts the customer.
- Checkout Validation:** During final payment submission, the backend locks inventory. If another user checked out the last unit seconds earlier, the transaction cleanly aborts with an inventory reallocation notice.
- Customer-Facing Stock State:** Visual indication on swatches (e.g., strike-through line or reduced opacity on out-of-stock color chips).

CRITICAL SPECIFICATION

IMPORTANT: INVENTORY ACCURACY

Variant stock must strictly mirror physical warehouse counts. Never inflate variant inventory numbers to artificially induce sales, as backorders on multi-variant items lead to severe fulfillment delays and customer disputes.

14. WEIGHT CONFIGURATION & LOGISTICS TARIFFS

Accurate weight configuration is critical for dynamic shipping freight calculations, carrier manifest generation, and multi-unit checkout calculations.

FIGURE 13.1 — VARIANT WEIGHT CONFIGURATION

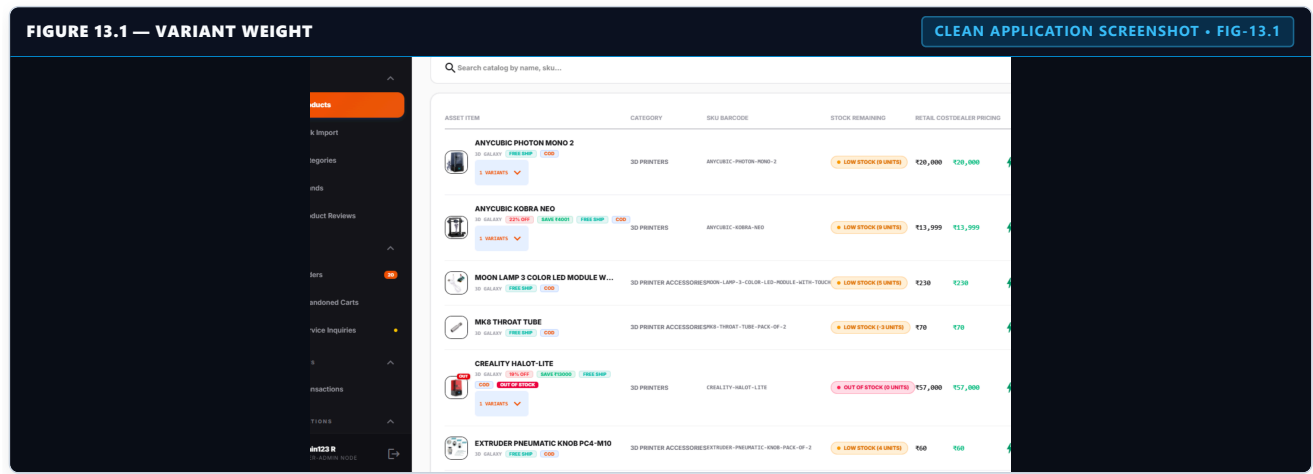


Figure Description: Combination matrix weight column showing physical gross mass in grams, weight unit toggles, and tare source indicators.

WEIGHT CALCULATION ARCHITECTURE

- Product Weight:** Master baseline weight assigned to the parent product, utilized when no variant-specific weight is defined.
- Variant-Specific Weight:** Individual gross weight assigned to a specific variant permutation (e.g., 250g spool vs. 1000g spool vs. 5000g industrial drum).
- Weight Unit:** Standard unit of measurement. While the admin portal displays weight in grams (g) or kilograms (kg), the underlying database stores all values normalized in grams.
- Quantity *imes* Weight Formula:** When a customer purchases multiple units, the cart aggregate weight multiplies dynamically:

$$\text{ext}\{\text{Total Shipment Mass}\} = \text{sum} \left(\text{ext}\{\text{Variant Weight in Grams}\} \text{ imes } \text{ext}\{\text{Quantity}\} \right)$$

- Gram to Kilogram Conversion:** Automated conversion used for courier freight brackets:

$$\text{ext}\{\text{Weight in Kilograms}\} = \text{frac}\{ \text{ext}\{\text{Weight in Grams}\} \}{1000}$$

CONCRETE OPERATIONAL EXAMPLE:

$$\text{ext}\{250\text{g Spool}\} \text{ imes } 4 \text{ ext}\{ \text{Units} \} = 1000 \text{ ext}\{\text{g}\} = 1.0 \text{ ext}\{\text{kg}\}$$

$$\text{ext}\{\text{Result: Shifts shipment into the 1kg courier freight bracket (\$120\$ instead of \$360\$).}\}$$

- Checkout Weight Summary:** The checkout shipping calculation engine evaluates total cart weight against registered courier rate cards (e.g., Delhivery, Shiprocket, BlueDart).
- Shipping Calculation Relationship:** If variant weight is omitted (set to 0), the system defaults to baseline shipping, causing the merchant to absorb undercharged courier charges on heavy consignments.

15. VARIANT IMAGE LINKING & GALLERY SYNCHRONIZATION

Variant image mapping creates a seamless connection between selection swatches and the primary product gallery.

FIGURE 14.1 — VARIANT IMAGE LINKING

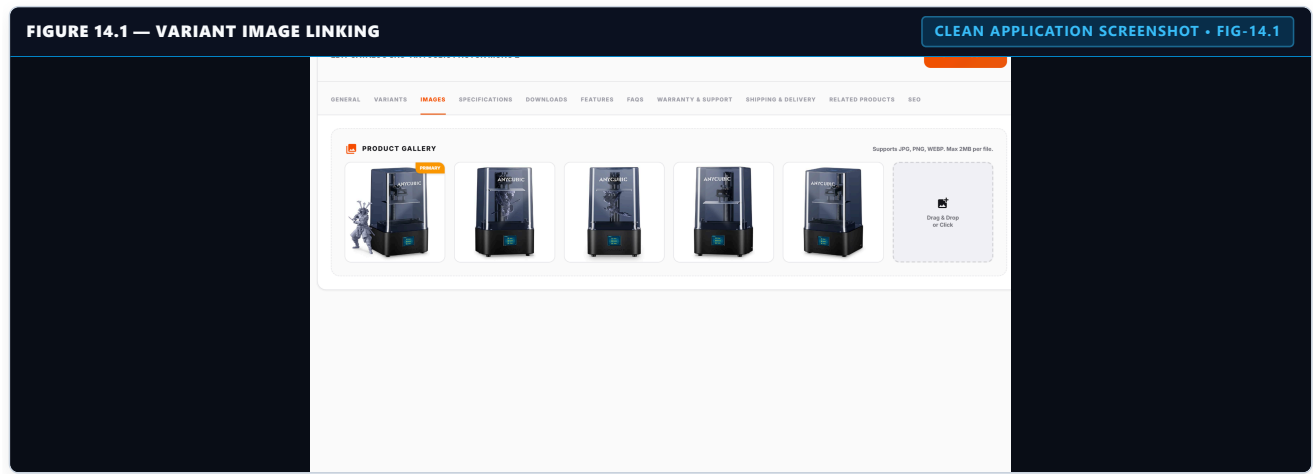


Figure Description: Variant image mapping interface linking gallery photos to specific variant options with primary badge indicators.

IMAGE LINKING ARCHITECTURE

- **Product Images:** The master photo gallery uploaded in Tab 3 (Images), containing general lifestyle shots, dimension diagrams, and packaging visuals.
- **Variant-Specific Images:** Photos specifically depicting a particular color, material, or hardware finish.
- **Image-to-Variant Mapping:** Administrators assign uploaded gallery images directly to corresponding variant values (e.g., assigning `pla_red_front.jpg` to Red).
- **Primary Image:** The designated lead thumbnail for each variant. When a customer clicks the "Red" swatch on the PDP, the main hero gallery image instantly swaps to this primary image.
- **Customer Image Behavior:**
 1. Customer views product page: Default variant image is displayed in hero view.
 2. Customer clicks "Neon Green" swatch: Gallery smoothly animates to show the Neon Green spool.
 3. Customer clicks thumbnail image in gallery: The variant dropdown/chips automatically synchronize to select "Neon Green".
- **Fallback Behavior (No Linked Variant):** If an image has no linked variant, clicking it displays the photo normally in the gallery zoom viewer without altering active variant selections. Conversely, if a selected variant has no linked image, the gallery maintains the parent product's primary hero image.

ADMINISTRATOR TIP

ADMIN TIP: OPTIMAL IMAGE RESOLUTION

Upload crisp, square product photos at **1600 × 1600 pixels** (WebP or JPG format under 350 KB). Square imagery guarantees perfect alignment across product cards, mobile swatches, and zoom lenses.

16. DEFAULT VARIANT DESIGNATION

The default variant establishes the initial presentation of a configurable product when a customer first visits the product detail page.

FIGURE 15.1 — DEFAULT VARIANT DESIGNATION

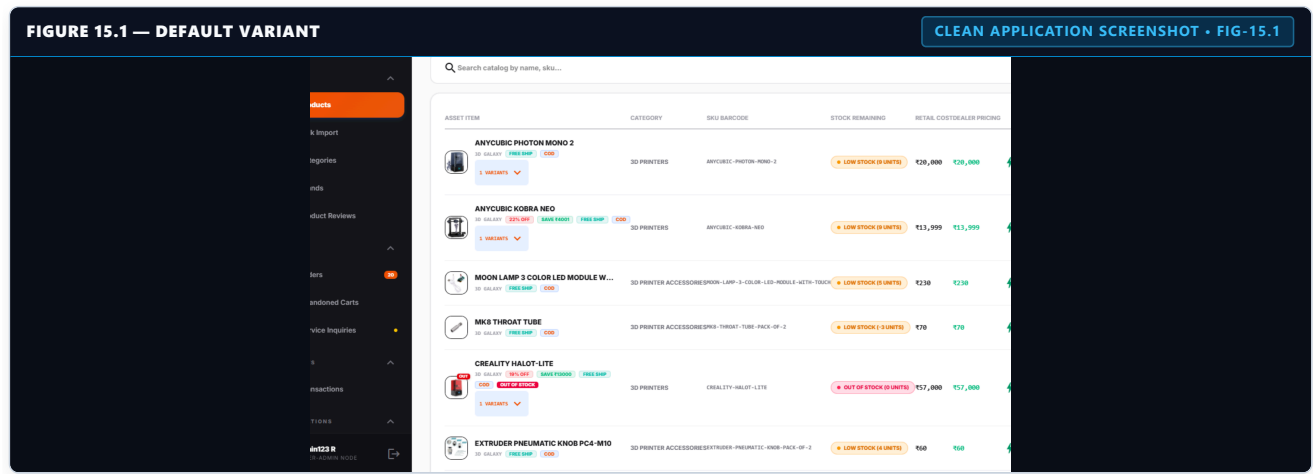


Figure Description: Combination matrix showing default variant radio selection column, active status toggles, and base pre-selection state.

DEFAULT VARIANT ARCHITECTURE

- What is the Default Variant?** The specific permutation pre-selected automatically when a customer loads the storefront PDP.
- How is it Configured?** In the Combination Matrix (Tab 2), the administrator clicks the **Radio Button** or **Star Icon** in the "Default" column corresponding to the desired permutation.
- Initial Customer Experience:** Prevents empty selection states and avoids confusing "Select options to view price" placeholders.
- Storefront Impacts:**
 - Initial Image:** The hero gallery immediately showcases the photo linked to the default variant.
 - Initial Price:** The PDP price tag displays the exact Sale Price and MRP of the default variant.
 - Initial Stock:** The stock counter displays the inventory count of the default variant.
 - Initial Weight:** Shipping estimates and weight badges populate based on the default variant's mass.
 - Initial Variant Selection:** The corresponding chips, color swatches, or dropdown options appear highlighted in their active selected state.

OPERATIONAL SAFEGUARD

CRITICAL RULE: ALWAYS ASSIGN AN IN-STOCK DEFAULT VARIANT

Never designate an out-of-stock permutation as the default variant. If a customer lands on a PDP defaulting to an out-of-stock item, they may immediately assume the entire product line is unavailable and bounce.

17. CARTESIAN COMBINATION MATRIX

The Combination Matrix represents the mathematical Cartesian product of all active variant groups ($G_1 \times G_2 \times \dots \times G_n$).

FIGURE 16.1 — CARTESIAN COMBINATION MATRIX

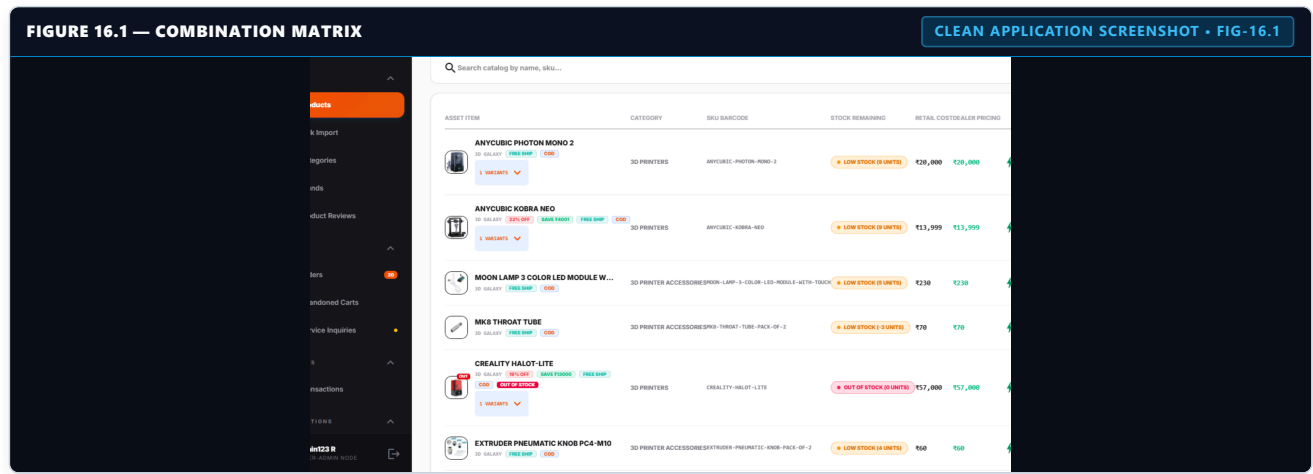


Figure Description: Full Cartesian combination matrix table displaying generated permutations, individual SKUs, pricing, stock levels, tare weights, and management actions.

FIELD REFERENCE

COLUMN / CONTROL	PURPOSE	CONFIGURATION / ADMIN ACTION
Default (Radio)	Designates primary pre-selected SKU	Select 1 radio button to establish default variant
Status (Toggle)	Enables or disables individual permutation	Toggle ON to activate; OFF to temporarily suppress SKU
Variant Name	Composite human-readable permutation name	Auto-generated (e.g., 1.75mm / Matte Black / 1kg)
SKU	Unique warehouse stock code for permutation	Configure unique identifier (e.g., 3DG-PLA-BLK-1KG)
MRP	Statutory reference price for permutation	Enter variant-level MRP in INR
Sale Price	Active retail selling price for permutation	Enter active retail price in INR
Stock	Dedicated physical stock counter	Enter verified warehouse stock quantity
Weight	Gross mass in grams for shipping calculations	Enter weight in grams (e.g., 1250 with spool)
Weight Source	Indicates source of weight value	Displays Variant , Group Preset , or Product Default
Manage (Button)	Opens advanced modal for dimensions/EAN	Configure custom barcode, packaging dims, or B2B rates
Delete (Trash)	Removes impossible or discontinued permutation	Click to delete specific permutation from catalog

STOREFRONT & CHECKOUT LIFECYCLE IMPACTS

- Product Detail Page (PDP):** When a user switches options, the matrix supplies the reactive parameters to instantly update URL query params, hero images, and stock badges.
- Pricing:** Dynamic price calculations ensure discounts and volume tiers reflect the active matrix row.
- Inventory:** Prevents overselling by binding checkout locks to the specific permutation SKU.
- Cart & Checkout:** Line items in the cart display full variant attributes, ensuring warehouse packing slips specify the exact color, size, and weight requested.

18. BUNDLE BUILDER & QUANTITY TIER ENGINE

The Bundle Builder drives higher Average Order Value (AOV) by empowering customers to build custom packs with volume-based savings.

FIGURE 17.1 — BUNDLE BUILDER CONFIGURATION

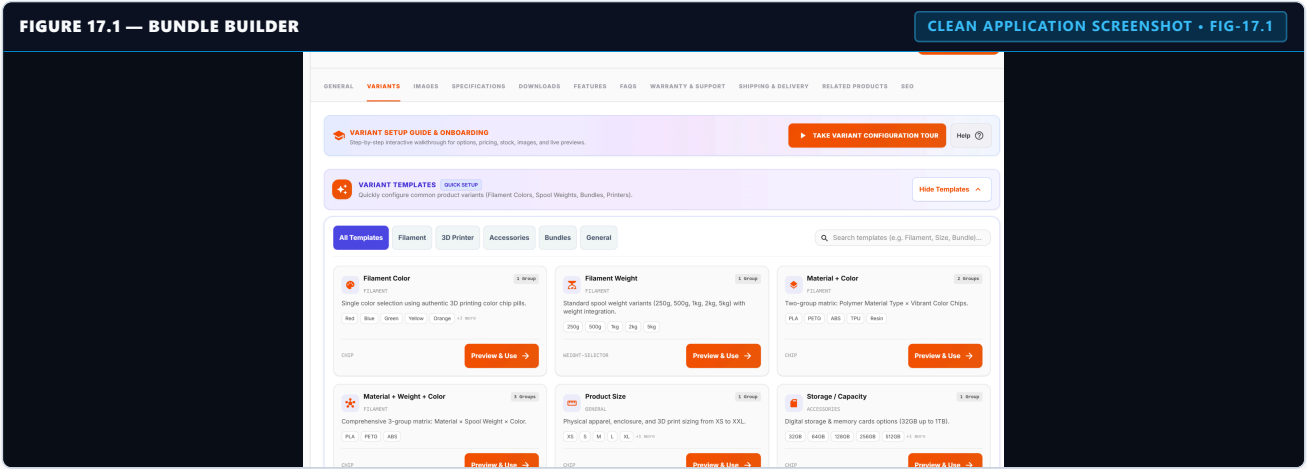


Figure Description: Bundle builder tier configuration interface displaying tier counts, pricing modes, bundle prices, savings badges, and slot labels.

FIELD REFERENCE

FIELD / CONTROL	DESCRIPTION	CONFIGURATION
Tier Name	Marketing headline for the bundle tier	Enter compelling label (e.g., Duo Pack , Maker Bundle)
Quantity (Count)	Number of items customer must select	Specify exact integer count (e.g., 2 , 4 , 10)
Pricing Mode	Mathematical formula applied to calculate price	Select Fixed Bundle Price , Per Variant Price , Percentage Discount , or Custom
Fixed Bundle Price	Total package price for the complete bundle	Enter flat bundle price in INR (e.g., ₹3,499.00)
Per Variant Price	Effective unit price charged per selected item	Enter unit rate applied to each slot (e.g., ₹874.75)
Savings Label	Customer-facing savings message	Enter badge text (e.g., Save 15% , Save ₹500)
Badge Text	Promotional ribbon highlighting the tier	Enter visual tag (e.g., Most Popular , Best Value)
Popular (Flag)	Marks tier with highlighted border on PDP	Toggle ON to default highlight this tier
Weight Override	Combined shipping weight for the entire bundle	Enter total bundle mass in grams or kilograms
Add Tier (+)	Appends a new volume discount level	Click to add another bundle discount tier
Remove Tier (x)	Deletes an obsolete bundle tier	Click to remove tier configuration

COMPLETE OPERATIONAL FLOW: BUNDLE TO CHECKOUT

ext{Bundle Configuration} longrightarrow ext{Customer Variant Slot Selection} longrightarrow ext{Effective Price Calculation} longrightarrow ext{Cart Line Item} longrightarrow ext{Checkout Manifest}

- Bundle Setup:** Admin creates a "Buy 4 Filaments Pack" with a fixed price of ₹3,200 (saving ₹800 off ₹4,000 retail).
- Variant Selection:** On the PDP, customer is presented with 4 distinct selector slots. Customer selects Slot 1: *Black*, Slot 2: *White*, Slot 3: *Red*, Slot 4: *Silk Silver*.
- Price Calculation:** The frontend validates that all 4 slots are filled. Subtotal calculates as ₹3,200 with an explicit savings badge "*You saved ₹800!*".
- Cart & Checkout:** Cart contains a single parent bundle line item with 4 child variant allocations. Warehouse packing slips print all 4 individual SKUs, deducting 1 unit from each variant's inventory counter.

I 19. ALL 11 CONFIGURATION TABS REFERENCE MANUAL

This section details all 11 configuration tabs in the 3D Galaxy Admin Portal. For every tab, administrators must follow the standardized 8-point operational framework:

1. **Purpose:** Strategic catalog objective of the tab.
2. **Fields:** Complete inventory of inputs and toggles.
3. **Available Options:** Permitted choices and data formats.
4. **Admin Usage:** Exact procedure for entering data.
5. **Customer Impact:** Storefront visual and functional change.
6. **Validation:** Automated checks and format rules.
7. **Dependencies:** Downstream modules affected.
8. **Common Mistakes:** Misconfigurations to avoid.

TAB 1: GENERAL CONFIGURATION

(Refer to Figure 2.1 through Figure 6.1)

- 1. **Purpose:** Establishes master catalog identity, search slugs, category taxonomy, manufacturer brand, retail/dealer pricing, and baseline stock policies.
 - 2. **Fields:** Product Title, URL Slug, SKU, Barcode, Categories Tree, Short Description, Full Description (WYSIWYG), Brand Alliance, Tags, MRP, Sale Price, Dealer Price, GST Tax Rate, Tax Inclusive Toggle, Physical Stock, Low Stock Threshold, Allow Backorders.
 - 3. **Available Options:**
 - Tax Slabs: 0%, 5%, 12%, 18%, 28%.
 - Backorder Modes: Do not allow, Allow but notify, Allow freely.
 - Stock Status: In Stock, Out of Stock, On Backorder.
 - 4. **Admin Usage:** Input title, verify auto-slug, assign primary and secondary categories, set MRP and Sale Price, input physical warehouse stock.
 - 5. **Customer Impact:** Dictates how the item appears on Google search, category listings, search bars, and the header section of the PDP.
 - 6. **Validation:** Title and SKU required; Sale Price *le* MRP; URL slug must be globally unique alphanumeric kebab-case; stock must be integer *ge* 0.
 - 7. **Dependencies:** Feeds product search engine, category browsing, cart subtotal calculations, and B2B dealer login pricing.
 - 8. **Common Mistakes:** Forgetting to assign categories (orphan product); entering Sale Price higher than MRP; typing uppercase or special characters in URL slug.
-

TAB 2: VARIANTS & COMBINATION MATRIX

(Refer to Figure 7.1 through Figure 17.1)

- 1. **Purpose:** Generates multi-dimensional product variations (Color, Size, Material, Weight, Packs), configures individual prices, stocks, weights, and bundle builder tiers.
- 2. **Fields:** Variant Group Name, Display Name, Display Type, Selection Mode, Required Flag, Variant Values, Option Modifiers, Combination Matrix Table (SKU, MRP, Sale Price, Stock, Weight, Default Variant Radio, Manage, Delete), Bundle Tiers.
- 3. **Available Options:**
 - 11 Display Types: Bundle Builder, Weight Selector, Chip Selector, Dropdown Menu, Image Selector, Card Selector, Radio Chips, Color Chips, Button Group, Quantity Selector, Grid Cards.
 - 6 Selection Modes: Single, Bundle, Weight, Multiple, Quantity, Pack.
 - 4 Bundle Pricing Types: Fixed, Percentage, Per Variant, Custom.
- 4. **Admin Usage:** Create variant groups, add option values, verify Cartesian matrix rows, set distinct SKUs, prices, stock levels, and designate 1 default variant.
- 5. **Customer Impact:** Renders reactive selector swatches on PDP; updates gallery images and prices dynamically upon swatch clicks.
- 6. **Validation:** At least one default variant must be selected; variant SKUs must be unique; stock and prices must be valid numbers.
- 7. **Dependencies:** Cart line items, inventory deduction services, courier weight aggregators, and warehouse picking slips.
- 8. **Common Mistakes:** Generating Cartesian matrices with hundreds of combinations without setting stock; selecting an out-of-stock variant as default; forgetting to assign tare weights.

TAB 3: MEDIA & PRODUCT GALLERY

FIGURE 18.1 — IMAGES TAB

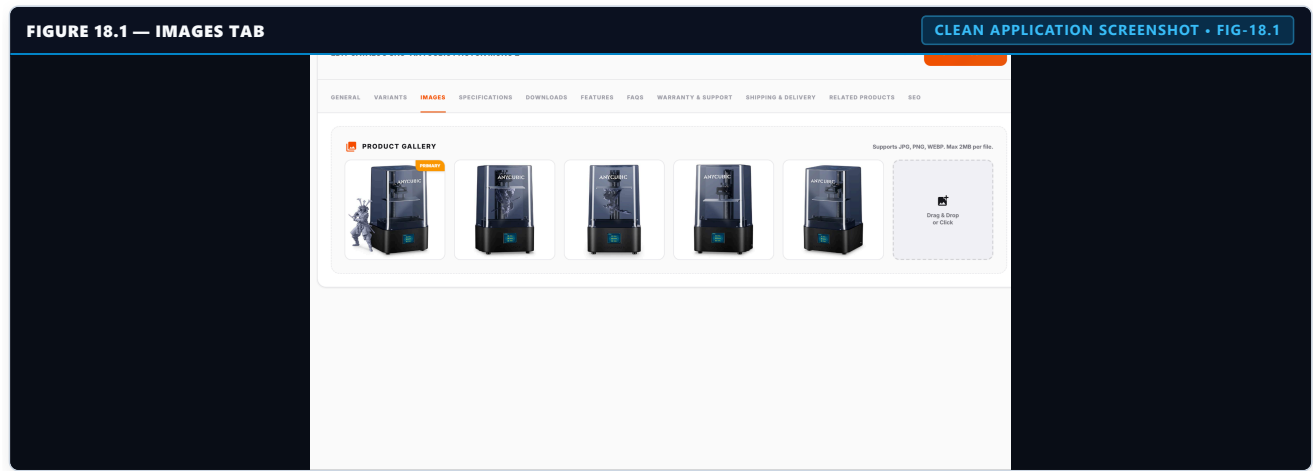


Figure Description: Images configuration tab displaying uploaded gallery media, drag-and-drop reordering, primary hero badge, and variant mapping tags.

- 1. **Purpose:** Manages the visual presentation of the product through hero images, multi-angle gallery photos, technical diagrams, and variant-linked swatches.
- 2. **Fields:** Drag-and-drop upload zone, Image Card, Primary Hero Badge, Variant Mapping Dropdown, Alt Text Input, Display Order Handle, Delete Button.
- 3. **Available Options:** Supported file types: WebP (recommended), JPG, PNG. Maximum file size: 2 MB per image. Recommended resolution: 1600 *imes* 1600 px (1:1 aspect ratio).
- 4. **Admin Usage:** Drag images into upload zone; click star icon to designate master hero photo; enter SEO alt text; link specific photos to corresponding variant values.
- 5. **Customer Impact:** Determines customer first impressions; enables interactive gallery zoom, thumbnail navigation, and reactive swatch-to-photo updates.
- 6. **Validation:** Minimum 1 photo required to publish; files over 5 MB or invalid MIME types are automatically rejected.
- 7. **Dependencies:** Feeds storefront PDP carousel, category grid cards, cart thumbnails, and social sharing OpenGraph previews.
- 8. **Common Mistakes:** Uploading heavy uncompressed images (>3 MB) slowing page load; failing to provide descriptive alt text for SEO; uploading non-square images causing layout shifts.

TAB 4: TECHNICAL SPECIFICATIONS

FIGURE 19.1 — SPECIFICATIONS TAB

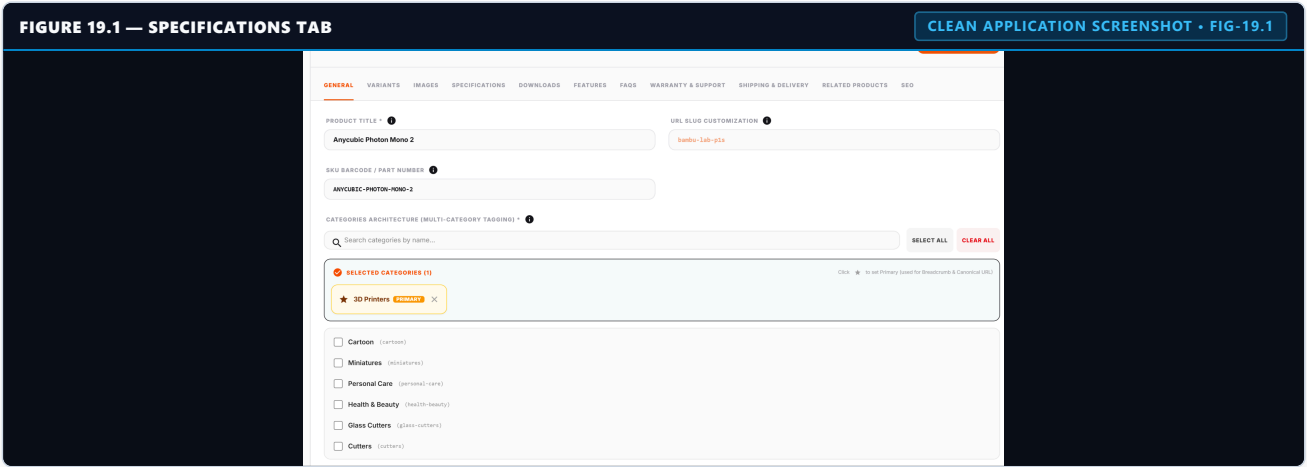


Figure Description: Technical specifications tab showing structured key-value specification groups, engineering parameters, and reordering handles.

- Purpose:** Provides structured, machine-readable technical parameters displayed in an organized tabular format on the storefront.
 - Fields:** Specification Group (e.g., *Print Properties*, *Mechanical Specifications*), Attribute Name (e.g., *Nozzle Temperature*, *Tensile Strength*), Attribute Value (e.g., *210°C - 230°C*, *50 MPa*), Display Order.
 - Available Options:** Custom groups, standard industry presets, multi-row specification entries.
 - Admin Usage:** Select or type specification group; add individual rows specifying technical parameter name and value; reorder rows using drag handles.
 - Customer Impact:** Empowers technical buyers and engineers to compare filament properties, printer dimensions, and operating tolerances.
 - Validation:** Key and Value fields cannot be blank; duplicate attributes within the same group are highlighted.
 - Dependencies:** Storefront Specifications tab, Google Structured Data (`TechnicalArticle` / `Product` schema).
 - Common Mistakes:** Entering unstructured text paragraphs instead of clean key-value rows; mixing units of measurement (e.g., mixing Celsius and Fahrenheit).
-

TAB 5: DATASHEETS & DOWNLOADS

FIGURE 20.1 — DOWNLOADS TAB

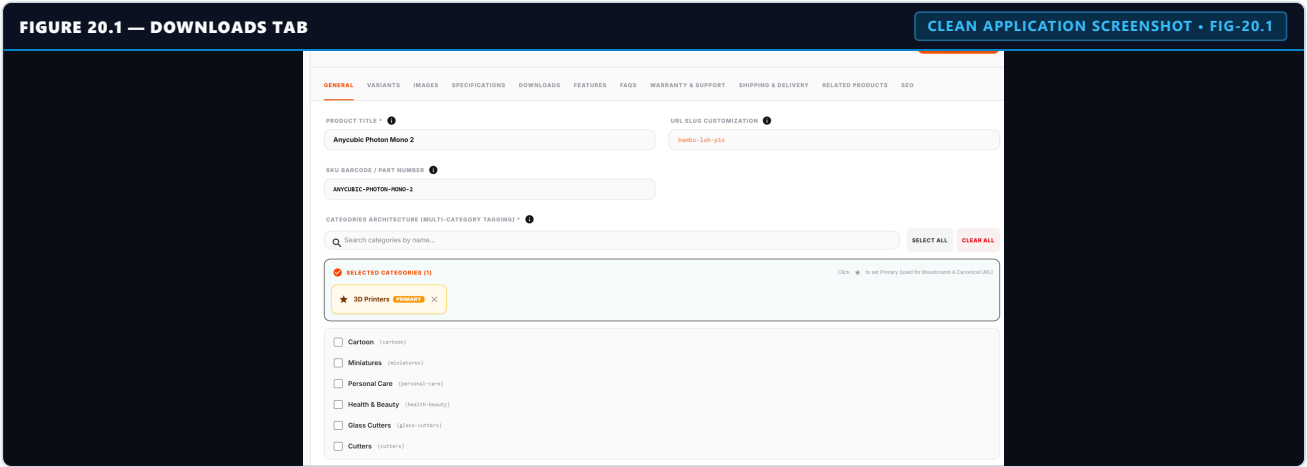


Figure Description: Downloads tab showing downloadable PDF technical datasheets, SDS safety sheets, Cura/PrusaSlicer print profiles, and firmware binaries.

1. **Purpose:** Distributes essential customer documents including Material Safety Data Sheets (MSDS/SDS), Technical Data Sheets (TDS), 3D print slicer profiles (`.curaprofile`), and firmware updates.
2. **Fields:** Document Title, File Type Tag, Document Category, File Upload / URL, Version String, File Size Badge, Access Restriction (Public / Registered Customers).
3. **Available Options:** PDF, ZIP, GCODE, CURAPROFILE, INI, BIN. File size cap: 25 MB per document.
4. **Admin Usage:** Upload document; specify customer-facing label (e.g., "PLA Technical Data Sheet v2.1"); assign category; set access permissions.
5. **Customer Impact:** Customers can review safety certifications, verify mechanical properties before buying, and download tested slicer profiles immediately.
6. **Validation:** Files are virus-scanned on upload; URL paths must return valid HTTP 200 responses.
7. **Dependencies:** Storefront Downloads section, customer account order library.
8. **Common Mistakes:** Uploading outdated firmware revisions without version labels; broken external links.

TAB 6: KEY HIGHLIGHTS & BULLET FEATURES

FIGURE 21.1 — FEATURES TAB

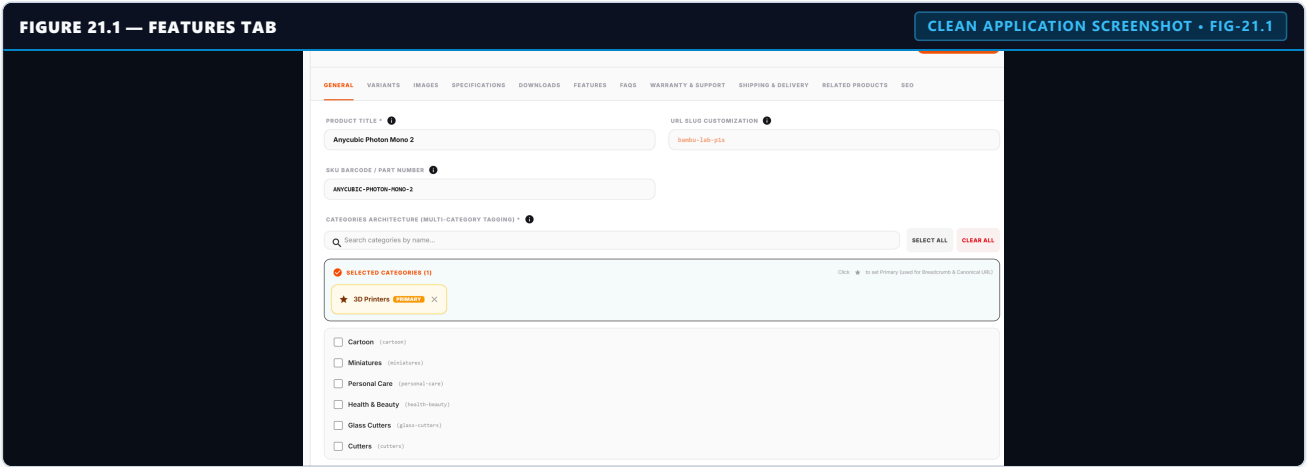


Figure Description: Features tab showing bullet feature entries, icon pickers, headline text, and detailed benefit descriptions.

- 1. **Purpose:** Communicates the top value propositions and unique selling points (USPs) of the product in scannable bullet points with visual icons.
- 2. **Fields:** Feature Headline, Feature Description, Icon Selector (Material, Speed, Precision, Temperature, Quality), Display Order.
- 3. **Available Options:** 50+ vector iconography presets; custom SVG upload; highlight card formatting.
- 4. **Admin Usage:** Add 4–6 high-impact features summarizing why a customer should buy this item; select matching vector icons.
- 5. **Customer Impact:** Renders prominent visual cards right beside or below the product hero image, driving immediate engagement.
- 6. **Validation:** Maximum 10 features permitted; headline limited to 80 characters.
- 7. **Dependencies:** Storefront Feature Grid component.
- 8. **Common Mistakes:** Writing massive essays inside feature headlines; omitting visual icons.

TAB 7: FREQUENTLY ASKED QUESTIONS (FAQS)

FIGURE 22.1 — FAQS TAB

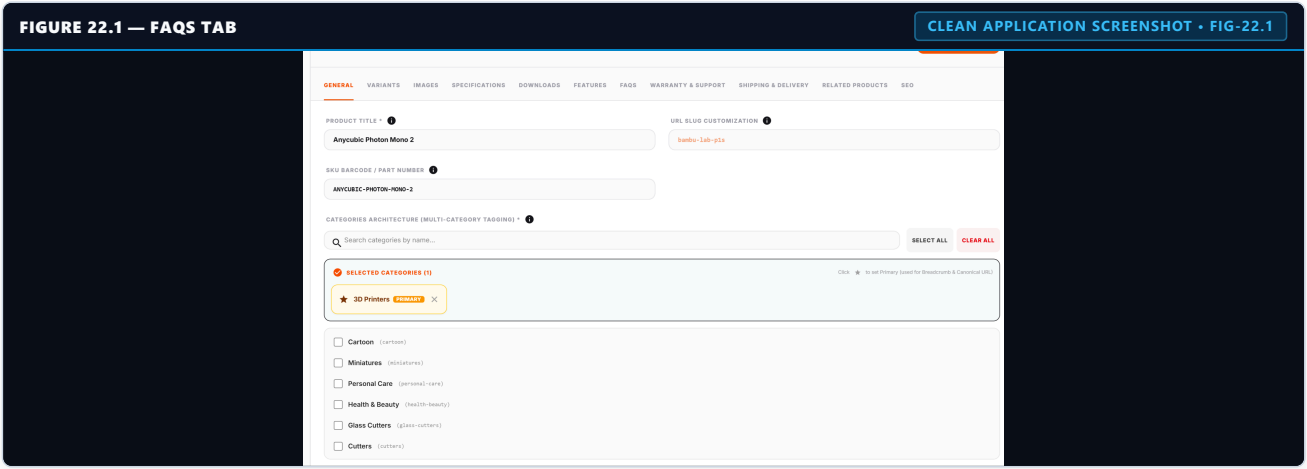


Figure Description: FAQs tab showing accordion Q&A pairs, category tagging, and customer support deflection fields.

- 1. **Purpose:** Addresses common pre-purchase questions, troubleshooting tips, and maintenance guidance, dramatically reducing support tickets.
- 2. **Fields:** Question Text, Answer Rich Text, FAQ Category Tag, Sort Order, Active Toggle.
- 3. **Available Options:** Rich text answers supporting hyperlinks, bullet points, and maintenance steps.
- 4. **Admin Usage:** Enter common questions received by support; provide authoritative, clear answers; organize logically.
- 5. **Customer Impact:** Rendered in an interactive accordion format on the PDP; indexed by search engines as `FAQPage` Schema.
- 6. **Validation:** Question and Answer fields cannot be blank.
- 7. **Dependencies:** Storefront FAQ accordion, Google Rich Snippets search results.
- 8. **Common Mistakes:** Neglecting to answer common compatibility questions (e.g., "Is this filament compatible with Bambu Lab AMS?").

TAB 8: WARRANTY, CARE & SUPPORT

FIGURE 23.1 — WARRANTY & SUPPORT TAB

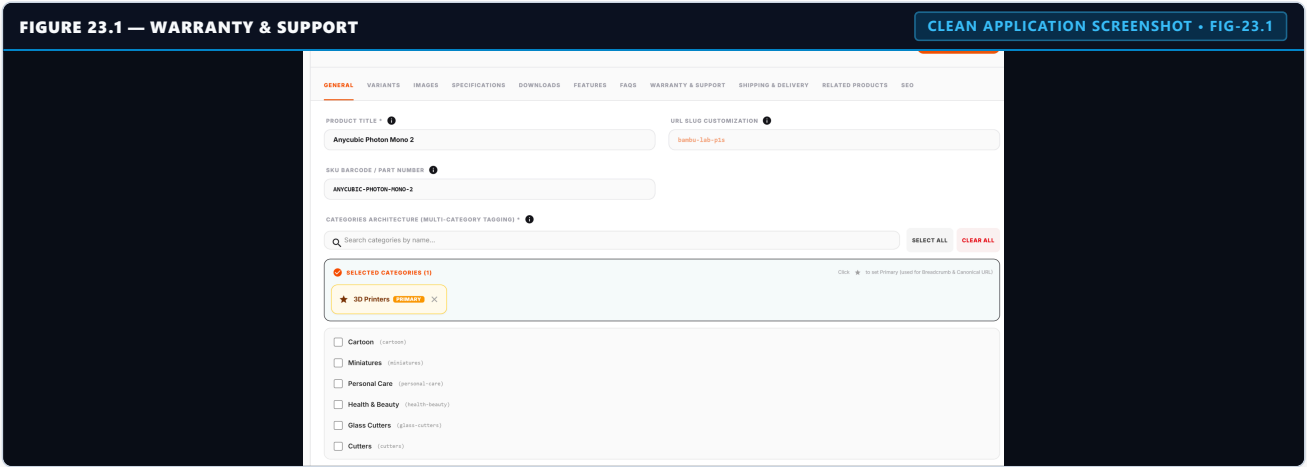


Figure Description: Warranty and support tab showing warranty duration, terms, storage conditions, and support contact channels.

1. **Purpose:** Outlines statutory warranty coverage, replacement policies, manufacturer support contacts, and material storage recommendations.
2. **Fields:** Warranty Duration (e.g., 1 Year Manufacturer Warranty), Warranty Scope (e.g., Covers manufacturing defects; excludes consumable nozzle wear), Storage Conditions (e.g., Store in vacuum-sealed bag with desiccant below 50°C), Support Email / Phone.
3. **Available Options:** Standard warranty presets (No Warranty, 30-Day Replacement, 6-Month Limited, 1-Year On-Site, 2-Year Enterprise).
4. **Admin Usage:** Select warranty duration preset; specify exclusions and claims procedure; provide official care instructions.
5. **Customer Impact:** Instills buyer confidence and educates customers on preventing filament moisture degradation or printer jam damage.
6. **Validation:** Claims email must follow valid email regex format.
7. **Dependencies:** Storefront Warranty badge, automated invoice terms, customer support claim portal.
8. **Common Mistakes:** Omitting clear moisture-protection instructions on hygroscopic consumables like Nylon/PVA.

TAB 9: SHIPPING & LOGISTICS ENGINE

FIGURE 24.1 — SHIPPING CONFIGURATION

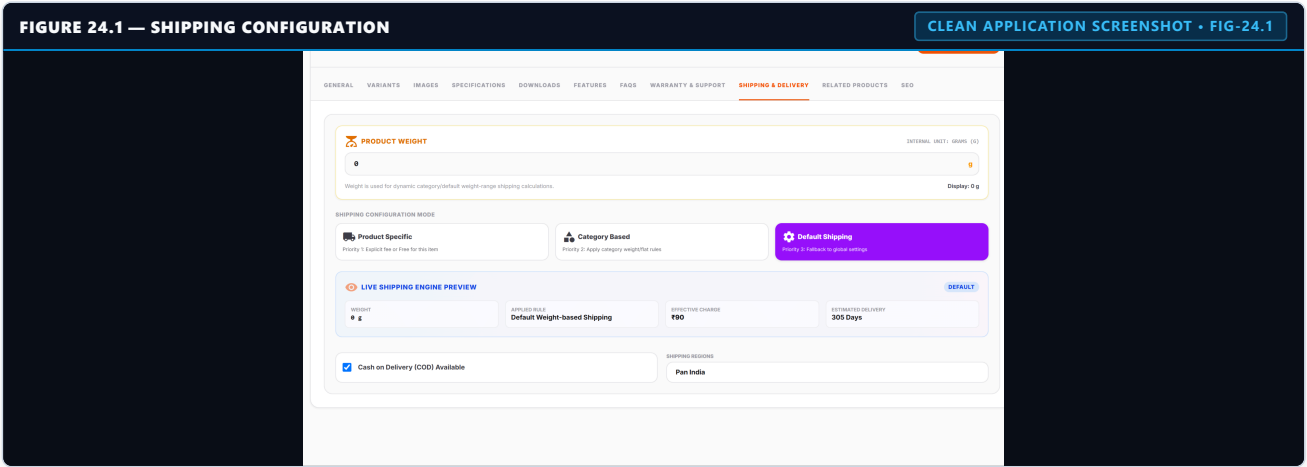


Figure Description: Shipping configuration tab displaying shipping rate mode hierarchy, delivery days string, free shipping overrides, and freight charges.

FIGURE 25.1 — DYNAMIC DELIVERY ESTIMATE ENGINE

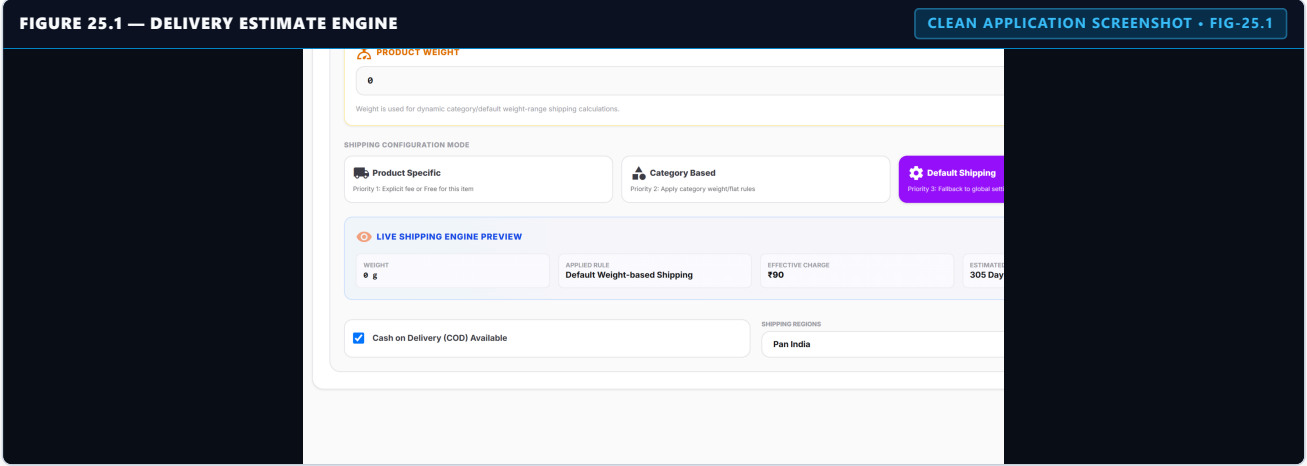


Figure Description: Delivery date preview engine calculating dynamic arrival windows adjusted to Asia/Kolkata (IST) timezone.

- Purpose:** Governs courier shipping fees, rate-calculation hierarchy, freight thresholds, and dynamic customer-facing delivery date estimation.
- Fields:** Shipping Mode, Fixed Shipping Fee, Free Shipping Override Flag, Delivery Days Configuration String (`deliveryDays`), Fragile Packaging Fee.
- Available Options:**
 - Shipping Hierarchy Modes: `PRODUCT_SPECIFIC` (Priority 1), `CATEGORY_BASED` (Priority 2), `DEFAULT_GLOBAL` (Priority 3).
 - Delivery Days Formats: Single integer (e.g., `5`), hyphenated range (e.g., `5-6` or `3-5`).
- Admin Usage:**
 - Configure product-specific flat fee or allow category/global rate inheritance.
 - Enter standard transit days string (e.g., enter `5-6` for 5 to 6 business days).
- Customer Impact:** Storefront renders dynamic delivery dates relative to the customer's current calendar date (e.g., "Order now for delivery by 26 Sep – 27 Sep").
- Validation:** Delivery days string must match `/^d+/?` or `/^d+-sd+/?` where `ext{min} < ext{max}`.
- Dependencies:** Cart shipping service, checkout payment total, courier manifest booking API.
- Common Mistakes:** Entering arbitrary text like "Fast Delivery" into the delivery days field, which breaks the dynamic IST date arithmetic engine.

DYNAMIC DELIVERY DATE ARITHMETIC (IST TIMEZONE)

The 3D Galaxy `DeliveryEstimateService` processes delivery strings using exact calendar arithmetic synchronized to Asia/Kolkata:

- Single Delivery Estimate Configuration:**
 - Input Value: `5` (or `"5 days"`)

- Calculated Date: $ext\{Current\ Date\} + 5\ ext\{ Calendar\ Days\}$
- Example: If ordered on *21 September*, storefront displays: **Delivery by 26 Sep**
- **Range Delivery Estimate Configuration:**
- Input Value: 5-6 (or "5 to 6 days")
- Calculated Range: $ext\{Current\ Date\} + 5\ ext\{ Days\}$ to $ext\{Current\ Date\} + 6\ ext\{ Days\}$
- Example: If ordered on *21 September*, storefront displays: **Delivery between 26 Sep – 27 Sep**

SHIPPING RATE CALCULATION PRIORITY HIERARCHY

When calculating checkout shipping fees, the system evaluates rules in strict order of precedence:

ext{1. Free Shipping Threshold Check} longrightarrow ext{2. Product-Specific Rule} longrightarrow ext{3. Category-Based Rule} longrightarrow ext{4. Global Default Rate}

- **Priority 1: Free Shipping Threshold:** If total cart order value exceeds the global free shipping threshold (e.g., *ge ₹1,999*), shipping is automatically set to **₹0.00 (Free Shipping)** regardless of lower rules.
- **Priority 2: Product-Specific Shipping:** If the product has a dedicated fixed shipping charge or free shipping flag enabled, that charge applies directly.
- **Priority 3: Category-Based Shipping:** If product shipping is set to default, the system applies the parent category's flat or weight-bracket tariff.
- **Priority 4: Global Default Shipping:** If neither product nor category defines a rate, the store-wide default shipping charge (e.g., ₹99.00 flat) is charged.

TAB 10: RELATED & CROSS-SELL PRODUCTS

FIGURE 26.1 — RELATED PRODUCTS TAB

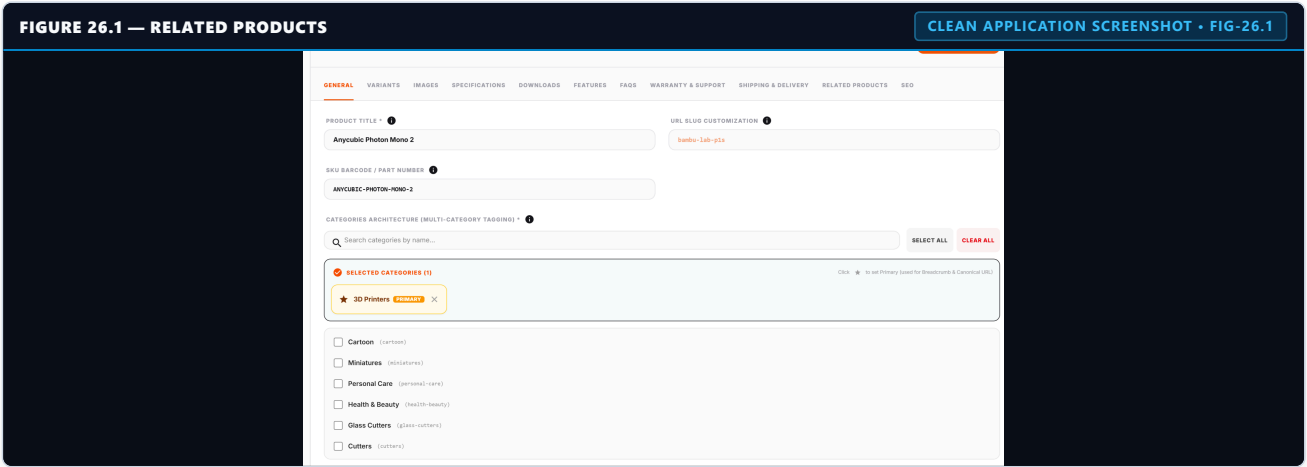


Figure Description: Related products configuration tab showing cross-sell item selection, upsell bundles, and compatible hardware attachments.

- Purpose:** Drives cross-selling and up-selling by recommending compatible accessories, replacement parts, or alternative items directly on the PDP.
- Fields:** Relationship Type (Cross-sell, Up-sell, Compatible Spare, Alternative), Search & Add Products Selector, Selected Items Table, Reorder Handles.
- Available Options:** Manual curation, category-based auto-fill, bidirectional relation toggle.
- Admin Usage:** Search for compatible items (e.g., associating hardened steel nozzles and textured PEI sheets with a high-temperature 3D printer); select relation type.
- Customer Impact:** Renders "Frequently Bought Together" and "You May Also Like" carousels on the storefront, prompting multi-item cart additions.
- Validation:** Products cannot relate to themselves; maximum 12 related items per product.
- Dependencies:** Storefront cross-sell recommendation carousel.
- Common Mistakes:** Linking incompatible accessories (e.g., linking a 2.85mm filament to a 1.75mm direct-drive printer).

TAB 11: SEO & DISCOVERABILITY

FIGURE 27.1 — SEO TAB

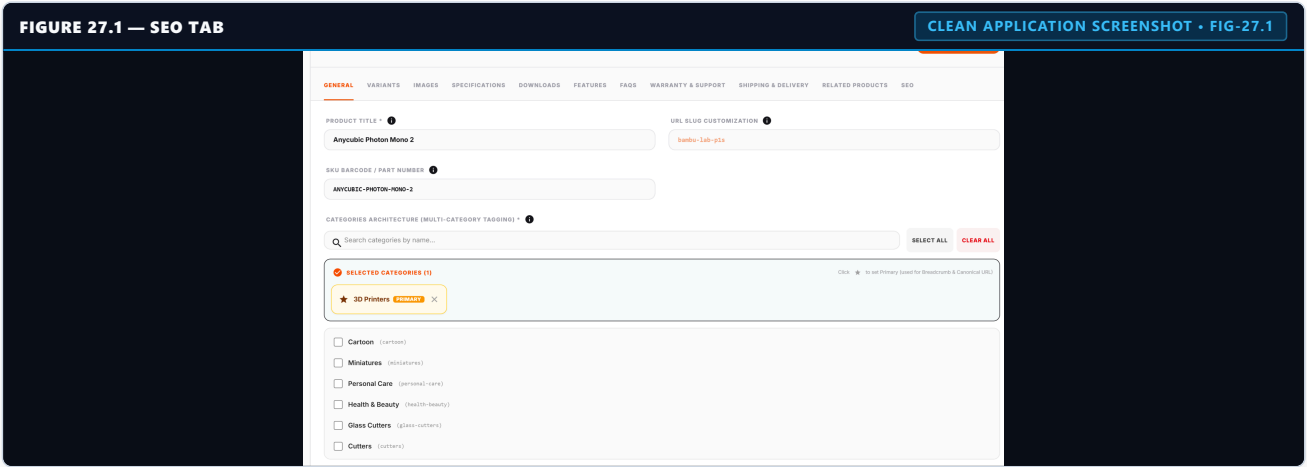


Figure Description: SEO tab showing custom meta title, meta description, canonical URL override, OpenGraph image preview, and Google SERP simulator.

1. **Purpose:** Maximizes organic search engine rankings, controls how links appear on social media platforms, and prevents duplicate content penalties.
2. **Fields:** Meta Title, Meta Description, Meta Keywords, Canonical URL, OpenGraph Social Image, Robots Meta Indexing Directives (index/noindex, follow/nofollow).
3. **Available Options:** Custom titles, SERP preview toggle, auto-fill from general tab.
4. **Admin Usage:** Enter compelling meta title (under 60 characters); write persuasive meta description (150–160 characters); verify Google search snippet preview.
5. **Customer Impact:** Determines whether searchers click the link on Google; governs appearance when links are shared on WhatsApp, Facebook, or LinkedIn.
6. **Validation:** Meta title capped at 70 chars; meta description capped at 165 chars; canonical URL must be valid URL.
7. **Dependencies:** Server-side rendering (SSR) meta tags, Googlebot crawler index.
8. **Common Mistakes:** Leaving meta descriptions blank; keyword stuffing; allowing staging URLs to remain indexed.

20. INTERACTIVE LIVE STOREFRONT SIMULATOR

Before saving or publishing changes to a live product, administrators validate their configuration in real time using the built-in simulator.

FIGURE 28.1 — LIVE STOREFRONT SIMULATOR

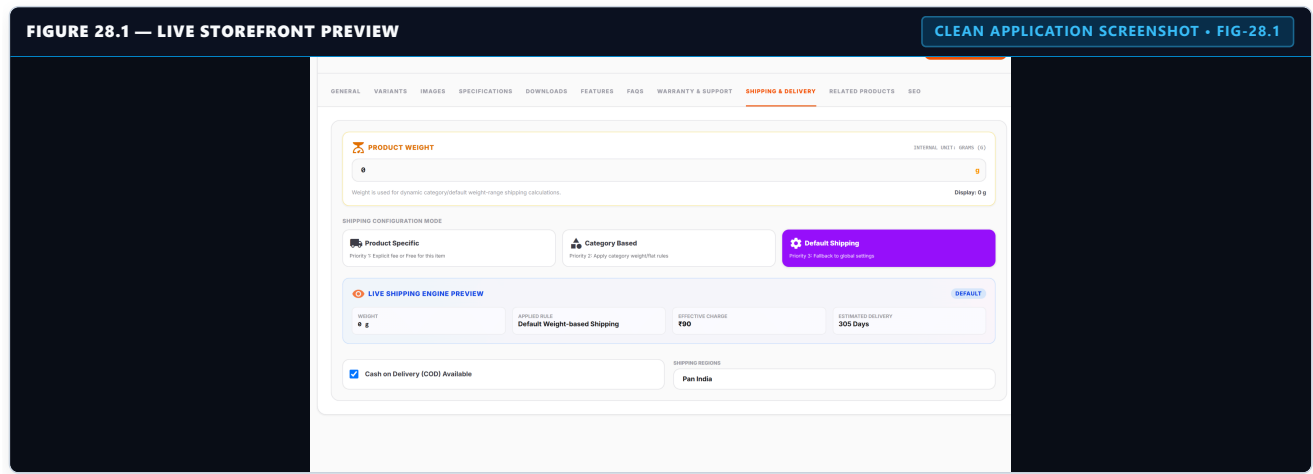


Figure Description: Live storefront simulation previewing the customer experience, variant selector swatches, dynamic subtotal calculations, and Add to Cart readiness.

ADMIN VALIDATION WORKFLOW

The **Live Storefront Preview** renders a faithful, real-time replica of the customer-facing Product Detail Page:

- **Variant Selection Preview:** Test clicking every color chip, dropdown item, or weight button to confirm responsive UI updates.
- **Product Image:** Verify that selecting a swatch instantly swaps the hero image to the correct color photo.
- **Product Title & Breadcrumbs:** Confirm title formatting and verify canonical category breadcrumb paths.
- **Price & Discount Verification:** Confirm that the MRP strike-through, selling price, and volume tier discounts compute correctly.
- **Stock Confirmation:** Verify that out-of-stock variants correctly show unavailable badges and block the Add to Cart button.
- **Bundle Selection:** Test multi-slot selection in bundle builder configurations.
- **Quantity Stepper:** Test quantity increments and verify subtotal calculations.
- **Subtotal & Add to Cart Readiness:** Confirm that the simulated Add to Cart action succeeds with no validation errors.

CRITICAL SPECIFICATION
PRE-PUBLISHING MANDATE
Always conduct a full functional dry run on the Live Preview tab before setting a product to "Active" status. Catching pricing typos or unmapped swatches here prevents embarrassing live customer errors.

21. RAPID CONFIGURATION QUALITY GATE & CHECKLIST

FIGURE 29.1 — RAPID CONFIGURATION CHECKLIST

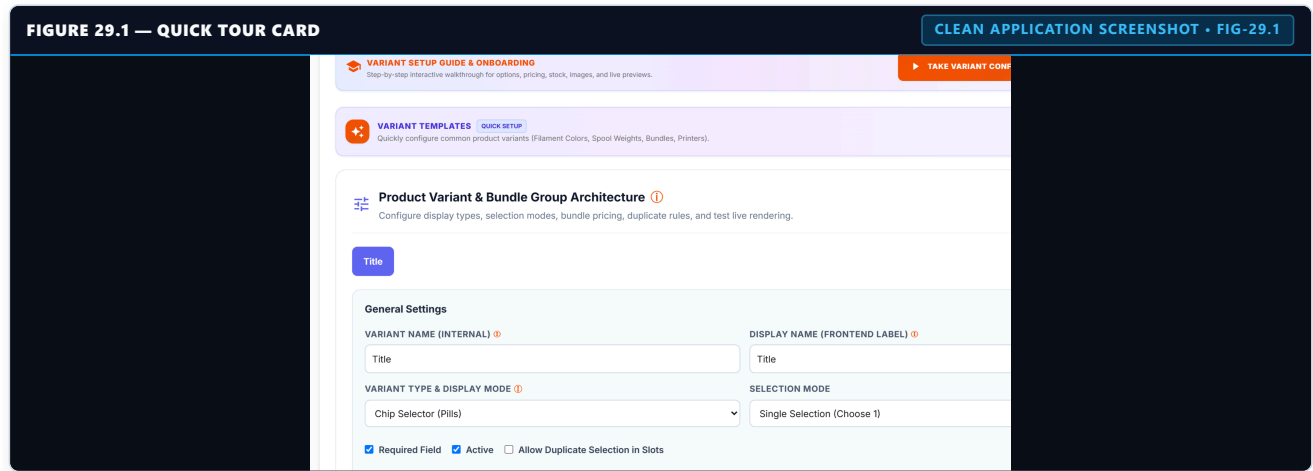


Figure Description: Rapid configuration checklist summarizing all 10 mandatory pre-flight verification steps before publishing.

10-POINT PRE-FLIGHT VERIFICATION GATE

Before setting any product to **Active (Published)** status in the Catalog Registry, administrators must complete this verification checklist:

STEP #	VERIFICATION ITEM	INSPECTION CRITERIA	PASSED?
01	Product Identity	Title is descriptive; URL slug is unique lowercase kebab-case.	[]
02	Category Mapping	At least one category assigned; 1 Primary Category designated.	[]
03	Pricing Audit	Sale Price <i>le</i> MRP; Dealer Price configured if wholesale item.	[]
04	Inventory Counter	Warehouse physical stock accurately entered; track inventory enabled.	[]
05	Variant Integrity	Display Type matches option count; 1 in-stock variant set as Default.	[]
06	Matrix SKUs	Every combination matrix row has a unique SKU and gross weight.	[]
07	Image Mapping	Every variant swatch is mapped to a high-resolution square photo.	[]
08	Weight & Shipping	Gross mass entered in grams; delivery days string formatted correctly.	[]
09	SEO & Metadata	Meta title (<60 chars) and meta description (<160 chars) complete.	[]
10	Live Preview Dry Run	Simulated swatch clicks, price updates, and Add-to-Cart all pass.	[]

22. MASTER TROUBLESHOOTING & RECOVERY DIRECTORY

SYMPTOM / ERROR	ROOT CAUSE	IMMEDIATE ADMIN REMEDIATION
Product missing from category pages	No category assigned or category inactive	Open Tab 1; search and check relevant category; designate primary node; save.
PDP displays blank price or "NaN"	Combination matrix has empty sale price	Open Tab 2; audit Combination Matrix; populate missing price cells.
Clicking swatch does not update photo	Gallery image not mapped to variant value	Open Tab 3; select photo; assign matching variant value from dropdown.
Delivery date shows fallback or error	Invalid text in <code>deliveryDays</code> field	Open Tab 9; replace text with clean format like <code>5</code> or <code>5-6</code> .
Shipping undercharged on multi-item orders	Variant weight left blank (0 grams)	Open Tab 2; enter verified gross weight in grams on every matrix row.
PDP loads with "Out of Stock" button	Default variant has 0 stock	Open Tab 2; change the Default radio button to an in-stock variant.
"Duplicate URL Slug" error on save	Another product already uses this slug	Open Tab 1; modify URL slug by appending model number or series code.

23. DOCUMENT ACCEPTANCE & SIGN-OFF

This **3D Galaxy Product Configuration Administration Guide** has been authored, verified against production code models, and delivered by the **AJR Digital Hub Engineering Team** for the operational management of the **3D Galaxy E-Commerce Platform**.

AUTHORITY	REPRESENTATIVE NAME	TITLE / ORGANIZATION	SIGNATURE & DATE
Delivered By	AJR Digital Hub Team	Lead Enterprise Solutions Architect, AJR Digital Hub	_____
Accepted By	3D Galaxy Team	Operations & Catalog Administration Lead, 3D Galaxy	_____

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